
Evaluating AI-Based Inventory Optimization from Traditional Methods to Neural Forecasting

Submitted by

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Confidentiality and Data-Provenance Statement

This study uses only public, non-proprietary data. The primary environment is the Walmart M5 competition dataset and the secondary environment is the Kaggle Store Item Demand Forecasting Challenge dataset; both are openly published benchmark datasets used under their communities' standard terms. No company-confidential, personal, or individually identifying data appear anywhere in this report, and the host-organisation dimension of the SIRP involves no client data.

Three commitments govern the evidence. First, provenance is explicit: every reported number derives from the frozen evidence package of the underlying research repository (number sheet, figure and table audits, claim traceability), generated programmatically rather than entered by hand. Second, results are reproducible: software seeds, manifests, run logs, and validation gates (44 PASS, 0 FAIL) are frozen so each step can be replayed by an independent auditor. Third, scope is stated honestly: the large-language-model phase of the research program was designed but not executed, and no LLM result is presented, implied, or charted in this report.

Executive Summary

Retail inventory management turns on two coupled decisions: how much customers will want, and how much stock to hold in anticipation. Most forecasting research addresses only the first. This study addresses both. Twelve forecasting approaches, from the moving average to a global long short-term memory (LSTM) neural network, are compared on two retail demand environments: 500 intermittent-demand series from the Walmart M5 data and all 500 series of a dense store-item demand dataset. Every model is evaluated under identical conditions: the same calendar window, a 28-day horizon over eight rolling origins, the same error metrics, and the same downstream inventory policy, so that forecasts are judged by the costs they generate as well as the errors they contain.

On sparse M5 demand, the LSTM records the lowest forecast error (mean absolute scaled error, MASE, of 1.316) and the lowest inventory cost (152.83), ranking first under 25 of 27 tested inventory policies. On dense store demand, the LSTM again records the lowest forecast error (MASE 0.978) but ranks fourth on inventory cost (2,247.46), behind a moving average (2,084.50); policy leadership there is fragmented across moving average, SES, naive, and DES (9, 7, 6, and 5 policies of 27, respectively). The implication is that lower forecast error does not necessarily produce lower inventory cost. Ninety-one paired statistical comparisons with Holm correction distinguish substantive differences from sampling variation, and a 27-policy sensitivity grid distinguishes conclusions that are stable across policies from ones that depend on them. All data, code, and validation gates are frozen and traceable. The large-language-model phase of this research program was designed but not executed and appears here only as future work.

Keywords: retail demand forecasting; inventory optimization; intermittent demand; M5 competition; LSTM; Croston method; forecast evaluation; MASE; order-up-to policy; sensitivity analysis.

	M5 (sparse, intermittent)	Store (dense, smooth)
Lowest forecast error	LSTM (MASE 1.316)	LSTM (MASE 0.978)
Lowest inventory cost	LSTM (cost 152.83)	Moving average (cost 2,084.50)
Stability	Stable: LSTM first in 25/27 policies	Policy-dependent: lead shared 9 / 7 / 6 / 5
Correspondence	Lowest error → lowest cost	Lowest error →/ lowest cost

Lower forecast error does not necessarily imply lower inventory cost.

Abbreviations and Definitions

Abbreviation	Expansion / Definition
ADI	Average Demand Interval
AI	Artificial Intelligence
ARIMA	AutoRegressive Integrated Moving Average
CI	Confidence Interval
CV^2	Squared Coefficient of Variation
DES	Double Exponential Smoothing (Holt's method)
DM	Diebold–Mariano (test)
EDA	Exploratory Data Analysis
H	Holding cost per unit per day (inventory policy)
$H1-H4$	Research hypotheses 1–4
L	Lead time in days (inventory policy)
LLM	Large Language Model
LSTM	Long Short-Term Memory
MA	Moving Average
MAE	Mean Absolute Error
MASE	Mean Absolute Scaled Error
M5	Walmart M5 Forecasting Competition dataset
P	Stockout (lost-sales) penalty per unit
P/H	Ratio of stockout penalty to holding cost
RMSE	Root Mean Squared Error
RMSSE	Root Mean Squared Scaled Error
RQ	Research Question
SARIMA	Seasonal ARIMA
SBA	Syntetos–Boylan Approximation
SES	Simple Exponential Smoothing
sMAPE	Symmetric Mean Absolute Percentage Error
TES	Triple Exponential Smoothing (Holt-Winters)
TSB	Teunter–Syntetos–Babai method
WAPE	Weighted Absolute Percentage Error

Introduction

Background

Retail inventory management turns on two coupled decisions: how much customers will want, and how much stock to hold in anticipation. Most forecasting research addresses only the first; this study addresses both. The empirical setting is retail replenishment, where every forecasting method is judged twice: once by the errors it makes, and once by the inventory cost its forecasts generate under a shared ordering policy.

Two demand environments frame the analysis. The first, captured by the Walmart M5 data, is sparse and intermittent: most items sell nothing on most days, with a 64.5% zero rate across the studied series. Classical forecasting tools were developed largely for smooth, continuous demand and are poorly suited to this setting. The second, a store-item demand dataset, is dense: nearly every item sells every day (zero rate 0.02%). A method that performs well in one environment may perform poorly in the other, so any comparative claim is tested in both.

Problem Context

Retail inventory decisions balance two costs. Ordering too much ties up cash in holding cost; ordering too little forfeits margin through stockouts. The preferred order quantity lies between these extremes, and locating it begins with demand prediction but does not end there. A forecast is an estimate; an order commits resources. This study covers the full path from estimate to committed cost.

Research Problem

Forecast accuracy and inventory cost weight errors differently. Consider two forecasters where A has lower average error than B. If A's errors concentrate in under-prediction on high-volume days while B's errors are modest over-predictions on quiet days, B can produce the lower inventory cost under a real ordering rule. Accuracy metrics treat each unit of error symmetrically, while inventory costs penalize stockouts and overstocks asymmetrically, filter forecasts through safety-stock formulas, and interact with lead times and review rhythms. Evaluating a forecasting model without passing its forecasts through an inventory policy therefore measures only part of its economic effect. Every model in this study is evaluated twice: once as a forecaster and once as an input to inventory decisions under a single shared ordering policy.

Rationale

The rationale for this study is practical and methodological at once. Procurement, replenishment, and engineering decisions are inventory decisions, and accuracy-only evidence, as the literature itself implies, is insufficient ground for them (Chapter 2). What is missing is a controlled design in which successive generations of forecasting methods face identical data, horizons, metrics, and one shared ordering policy on both sparse and dense retail demand, so that the forecast-to-decision relationship

can be measured rather than assumed. This project supplies that design and freezes its evidence for independent audit.

Research Objectives

1. Compare twelve forecasting approaches (baselines, exponential smoothing, ARIMA/SARIMA, Croston-family specialists, and a global LSTM) on identical footing across two demand environments.
2. Evaluate each model on forecast accuracy and on the inventory cost its forecasts generate under a shared ordering policy.
3. Distinguish statistical significance from practical importance using paired tests, effect sizes, and confidence intervals.
4. Distinguish policy-stable conclusions from policy-sensitive ones using a 27-policy sensitivity grid.
5. Publish a frozen, traceable evidence base that supports independent audit of every reported number.

Research Questions

How does the effectiveness of AI-based inventory optimization change as we move from traditional forecasting models toward neural-network approaches, when effectiveness is measured at the level of inventory outcomes rather than forecast errors alone?

Hypotheses

Forecast accuracy differs significantly across model generations on the same series and horizons.

Forecast-accuracy differences translate into measurable inventory-outcome differences under a common policy.

Increasing model sophistication yields monotonically improving inventory outcomes. (The claim under test; rejection is a publishable finding, not a failure.)

Observed patterns are consistent across the two retail environments.

Verdicts appear in section 5.3.

Scope

The study covers univariate daily demand forecasting (no prices, promotions, or calendar features), a 28-day horizon over eight rolling origins, and one family of inventory policies (daily-review order-up-to, lost sales). The M5 analysis uses a stratified sample of 500 series rather than the full 30,490 catalog. SARIMA runs Store-only and the Croston family M5-only, each for documented methodological reasons. The planned large-language-model phase was designed but not executed; it appears only as future work (section 6.6).

Significance and Contribution

The empirical contribution is a measurement of the forecast-decision relationship in both directions within one controlled design: correspondence on M5 demand and divergence on Store demand. The methodological contribution comprises a shared-policy evaluation harness that holds the inventory treatment constant across models, a leakage-safe rolling protocol with scale-free metrics, and a sensitivity discipline in which headline rankings are reported together with their stability across policies. The analytical contribution pairs each significance statement with an effect size and each headline ranking with a policy-stability assessment. For reproducibility, the frozen evidence package (number sheet, figure and table audits, claim traceability) maps every reported number to its source file.

Chapter Structure

Chapter 2 reviews the literature and locates the research gap. Chapter 3 sets out the methodology: data, series selection, the model ladder, evaluation metrics, statistical testing, the inventory policy, and the sensitivity design. Chapter 4 reports the analysis: exploratory demand structure, forecast boards, statistical comparisons, inventory results, and sensitivity. Chapter 5 consolidates the findings, evaluates the hypotheses, and states the recommendations. Chapter 6 concludes with limitations and future scope. Annexures A to J carry the detailed evidence: dataset and protocol details, model configurations, statistical output, pairwise results, sensitivity matrices, convergence and validation, reproducibility, and supporting tables and figures.

Literature Review and Research Gap

Review Strategy

The literature relevant to AI-based inventory optimization spans five decades and two disciplines: forecasting methodology and inventory control. The review was conducted backwards from the needs of the present design: the study requires a traditional statistical toolkit to benchmark, specialist methods for intermittent demand, a neural method for pooled learning, trustworthy accuracy measurement, and evidence on how forecasts behave once passed through stock control. Twelve peer-reviewed papers and books were selected as the load-bearing sources: foundational (Box and Jenkins), intermittent-demand specialists (Croston; Syntetos and Boylan; Syntetos, Boylan and Croston; Teunter, Syntetos and Babai), machine learning (Hochreiter and Schmidhuber), retail-scale competition evidence (Makridakis et al.), accuracy measurement (Hyndman and Koehler), comparison methodology (Diebold and Mariano; Holm), applied canon (Hyndman and Athanasopoulos), and a decision-level case study (Syntetos, Babai, Davies and Stephenson). Each is summarized with its method, context, key finding, relevance to this study, and limitation in Section 2.8. What is known is considerable — how to smooth, how to model autocorrelation, how to handle intermittency, how to compare accuracy. What is missing, and what motivates the present study, is systematic evidence on whether successive generations of forecasting machinery produce successively better inventory outcomes when the downstream policy is held constant.

Retail Demand Forecasting

Retail demand forecasting is the discipline of predicting item-level sales over short horizons, and its modern benchmark is the M5 competition (Makridakis et al. 2022): roughly 42,840 hierarchical Walmart daily series with intermittency, exogenous and pricing variables, evaluated jointly on point and uncertainty tracks. Companion results showed machine-learning methods exploiting grouped and exogenous data winning at retail scale. M5 thereby supplies both the benchmark dataset lineage this study draws on and the existence proof that pooled learning can beat per-series statistics on store data. The practitioner setting is defined by demand structure: in sparse, intermittent catalogs most items sell nothing on most days; in dense catalogs nearly every item sells every day. A method that performs well in one environment may perform poorly in the other, so comparative claims must be tested in both — the design principle adopted in Chapter 3.

Traditional Forecasting Methods

The modern toolkit begins with Box and Jenkins (Box and Jenkins 1970), whose *Time Series Analysis: Forecasting and Control* established the ARIMA identification—estimation—checking cycle for discrete time series. Their claim — that models built this way yield optimal forecasts and optimal control action — set the template for statistical forecasting: structure the dynamics, estimate from history, verify on residuals. Hyndman and Athanasopoulos (Hyndman and Athanasopoulos 2018), in *Forecasting: Principles and Practice* (2nd ed.), consolidated the applied canon — exponential smoothing (SES, Holt, Holt-Winters), ARIMA, dynamic regression — with the insistence on graphics, real business examples, and implementable software that made these methods the retail default. The practitioner baseline beneath all of this is simpler still: the naive carry-forward, the seasonal

naive, and the moving average — a low-pass filter on noisy demand with no estimated dynamics to break. The literature treats these not as straw men but as thresholds any method must clear, a discipline this study adopts directly in its model ladder.

Intermittent Demand Forecasting

Classical tools assume smooth, continuous demand. Croston (Croston 1972) showed what happens when that assumption meets intermittent demand: exponential smoothing “almost always produce[s] inappropriate stock levels,” with constant demand at fixed intervals generating up to double the stock needed. His remedy — smoothing nonzero demand sizes and inter-demand intervals separately, with adjusted safety-stock rules — founded the specialist literature. Syntetos and Boylan (Syntetos and Boylan 2005) demonstrated on 3,000 real automotive series that Croston’s estimator is biased upward, and proposed the bias-corrected Syntetos–Boylan approximation (SBA), which proved superior out of sample. Teunter, Syntetos and Babai (Teunter et al. 2011) went further: because Croston and SBA update only on positive demand, they cannot decay toward zero under obsolescence. Their Teunter–Syntetos–Babai (TSB) estimator updates demand probability every period, linking forecasting explicitly to inventory obsolescence. The progression is cumulative — bias identified, bias corrected, dynamics adapted — and each step was validated on real intermittent series, not simulations alone. Syntetos, Boylan and Croston (Syntetos et al. 2005) then replaced arbitrary industry cut-offs for “erratic” or “lumpy” demand with theory-based regions expressed through average inter-demand interval (ADI) and the squared coefficient of variation of demand sizes (CV^2), validated on the same 3,000-series base. Their categorization is the direct ancestor of the archetype analysis used in this study.

Machine Learning and Neural Forecasting

Hochreiter and Schmidhuber (Hochreiter and Schmidhuber 1997) diagnosed why recurrent networks failed on long sequences — vanishing error flow — and introduced long short-term memory (LSTM), with constant-error carousels and gating that bridge time lags beyond a thousand steps. The paper claims sequence memory, not retail forecasting; its relevance here is derivative but direct: LSTMs can learn weekly structure, regime shifts, and cross-series patterns that per-series linear models cannot express. Section 2.2 records how that possibility became retail reality in the M5 competition.

Inventory Optimisation and the Forecast-to-Decision Link

The critical turn comes from studies that pass forecasts through stock control. Croston (Croston 1972) already framed the problem as stock levels, not errors. Teunter, Syntetos and Babai (Teunter et al. 2011) tied estimator choice to obsolescence cost. Most directly, Syntetos, Babai, Davies and Stephenson (Syntetos et al. 2010), in a UK wholesaler case study, examined forecasting and stock control jointly and found “considerable scope for improving current practices,” with service-level versus cost trade-offs and managerial barriers deciding outcomes as much as forecast quality.

Forecast Accuracy versus Inventory Decision Quality

The distinction is structural. Accuracy metrics are symmetric, per-unit, and policy-free; inventory costs are asymmetric (a stockout costs multiples of a holding unit), thresholded through safety-stock formulas, and defined by the policy’s lead time, service target, and review rhythm. A sharper

forecaster that occasionally under-predicts can therefore cost more than a duller one that rarely does. The literature establishes each half of this mechanism — the metrics, the policies, the linkage cases — but rarely measures both halves for successive model generations under one held-constant policy. That is the gap.

Review of Twelve Relevant Research Papers

Table 2.1 compresses the twelve studies: method, context, key finding, relevance to the present design, and the limitation each leaves open.

Table 2.1: Literature synthesis: twelve studies — topic, methodology, key finding, relevance, and limitation/gap.

No.	Study	Method	Context	Key finding	Relevance	Limitation / gap
1	Box & Jenkins (1970), book	ARIMA identification—estimation—checking	Industrial / economic series	Optimal forecasts and control from structured dynamics	ARIMA rung; statistical baseline	Univariate, linear; no intermittency treatment
2	Croston (1972), OR Quarterly	Size/interval smoothing + safety-stock rules	Intermittent demand	SES on intermittent demand overstocks; separate the sequences	Croston rung; stock-level framing	Biased estimator; no demand-probability updating
3	Syntetos & Boylan (2005), IJF	SBA bias correction; 3,000 automotive series	Intermittent demand	SBA superior out of sample; bias matters	SBA rung; archetype tradition	Correction assumes stable size/interval processes
4	Teunter, Syntetos & Babai (2011), EJOR	TSB probability updating	Intermittent demand + obsolescence	TSB adapts faster; links forecasting to obsolescence	TSB rung; adaptivity hypothesis	Largely simulation-validated; parameter sensitivity
5	Syntetos, Boylan & Croston (2005), JORS	ADI-CV ² categorization; 3,000 series	Demand-pattern regimes	Theory-based SKU segmentation rules	Archetype analysis design	Categorization only; no policy-constant comparison
6	Hyndman & Koehler (2006), IJF	Scale-free accuracy theory	Competition / management series	MASE as cross-series standard	Headline metric ($m = 7$)	Accuracy metric; silent on decision cost
7	Hochreiter & Schmidhuber (1997), Neural Comp.	LSTM gating/memory	Long sequences	Bridges 1,000+ step lags	Global LSTM rung rationale	Not a forecasting or retail study
8	Makridakis et	M5	~42,840	ML wins at	Dataset lineage;	Accuracy-

No.	Study	Method	Context	Key finding	Relevance	Limitation / gap
	al. (2022), IJF	competition design	Walmart series	retail scale with grouped data	pooled-learning prior	focused; no held-constant inventory evaluation
9	Hyndman & Athanasopoulos (2018), OTexts	ETS/ARIMA applied canon	Business series	Implementable retail forecasting reference	SES/DES/TES/ARIMA configurations	Textbook; limited intermittent-specialist depth
10	Diebold & Mariano (1995), JBES	Equal-accuracy testing	General loss functions	Significant $\neq$ better; any loss allowed	91-pair inference backbone	Paired-design dependence inherited
11	Holm (1979), Scand. J. Statist.	Step-down multiplicity control	General	Family-wise error control with power over Bonferroni	Holm correction across 91 pairs	Control can be conservative under dependence
12	Syntetos et al. (2010), IJPE	UK wholesaler case study	Distribution inventory	Forecasting and stock control jointly improvable	Decision-quality precedent; gap anchor	Single case; no model-generation comparison

Thematic Synthesis

Read together, the twelve studies form four themes. First, the traditional toolkit (Box and Jenkins; Hyndman and Athanasopoulos) supplies estimable structure — smoothing, autoregression, seasonality — and the baselines any method must clear. Second, the intermittent-demand stream (Croston; Syntetos and Boylan; Syntetos, Boylan and Croston; Teunter, Syntetos and Babai) shows that the toolkit’s smooth-demand assumptions fail predictably under zeros, and that specialist estimators and theory-based demand categorization recover part of the loss. Third, the machine-learning stream (Hochreiter and Schmidhuber; Makridakis et al.) establishes that pooled, nonlinear learning can win at retail scale. Fourth, the evaluation stream (Hyndman and Koehler; Diebold and Mariano; Holm) disciplines how gains are measured: scale-free metrics, paired testing, and multiplicity control arrive as a package. Across all four themes the decision-level evidence is thin: only Syntetos et al. (2010) examines forecasting and stock control jointly, and none compares successive model generations under one held-constant inventory policy.

Research Gap

What is known: smoothing, ARIMA, intermittent-demand specialists, LSTMs, scale-free accuracy measurement, paired testing with multiplicity control, and case-level evidence that forecasting and stock control interact. What has been tried: each generation against accuracy leaderboards, and intermittency specialists against inventory cost in narrow settings. Where the literature disagrees or stays silent: whether increasing model sophistication improves inventory outcomes monotonically; whether accuracy leadership transfers into cost leadership across demand structures; and which

conclusions survive the policy grid. What is missing: a single controlled design in which traditional, statistical, intermittent-specialist, and neural forecasters face identical windows, horizons, metrics, and one shared ordering policy on both sparse and dense retail demand — with significance, effect size, and policy-stability reported together. Why this research is necessary: procurement and engineering decisions are inventory decisions, and accuracy-only evidence, as the literature itself implies, is insufficient ground for them.

Conceptual / Research Framework

The conceptual framework joins the four themes into one pipeline. Forecasting models, ordered by complexity (the model ladder of Chapter 3), produce 28-day forecasts under a leakage-safe rolling protocol. Forecasts are scored twice: as forecasts, by scale-free and absolute metrics with paired statistical testing; and as decisions, by passing every forecast through one shared inventory policy whose parameters define the effective loss function. Stability of the resulting rankings is then tested across a 27-policy grid. Four hypotheses ride on this framework: H1 on accuracy differences (traditional versus neural themes), H2 on accuracy-to-inventory translation (the forecast-to-decision link), H3 on monotone sophistication (the ladder ordering), and H4 on cross-dataset consistency (the two demand environments).

Link Between Literature Gap, Research Questions and Hypotheses

The research question asks how effectiveness changes as forecasting moves from traditional to neural methods when effectiveness is measured at inventory level. The gap of Section 2.10 supplies its parts: “successive model generations” come from the first three themes; “measured at inventory level” comes from the forecast-to-decision link; “held-constant policy” comes from the evaluation stream’s insistence on controlled comparison. The hypotheses operationalize the question: H1 tests whether generations differ in accuracy at all; H2 tests whether those differences move inventory outcomes; H3 makes the monotone-improvement claim explicit so that its rejection is informative; and H4 tests whether any of this transfers between a sparse and a dense demand environment. Verdicts are reported in Section 5.3.

Research Methodology

Research Design

The study is organized as a notebook sequence: a research map (00) states the question; environment control (01) fixes seeds and software; acquisition and audit notebooks (02a, 02b) verify both datasets; exploratory analysis (03) characterizes demand before model selection; experimental design (04) locks the protocol; a teaching notebook (05a) sets out time-series vocabulary; then the model rungs run in order (baselines in 05b, exponential smoothing in 06, Croston specialists in 06c, ARIMA/SARIMA in 07, LSTM in 08). A subsequent hardening pass corrected calendar, validation, comparability, metric, and policy issues and froze the evidence; see section 3.14. Each notebook's role in the report is recorded in the traceability matrix (`traceability/notebook_traceability.csv`).

Data Sources

Both datasets are public. The primary environment is the Walmart M5 competition data (Makridakis et al. 2022): thousands of item-store daily series, 2011 to 2016, studied here over the common window 2013-01-01 to 2016-05-22. The secondary robustness environment is the Kaggle Store Item Demand Forecasting Challenge: 10 stores by 50 items (500 series), a complete panel of 913,000 rows from 2013-01-01 to 2017-12-31, analyzed over the same common window. No proprietary, personal, or confidential data are used anywhere in the study.

Dataset Characteristics

The two environments differ sharply in demand structure: M5 is sparse and intermittent, with most items selling nothing on most days; Store Item Demand is dense, with essentially no zero-demand days and a weekly rhythm. Detailed demand characteristics and representative series appear in Sections 4.2 and 4.3; dataset and protocol specifics are frozen in Annexure A.

Sampling / Series Selection

Store demand uses all 500 series, so no sampling was required. M5 contributes a stratified sample of 500 (recorded in `05_experiments/m5_series_selection.json`) covering all demand archetypes and departments. Random sampling risked omitting rare but substantively important demand shapes, while full-catalog deep learning was computationally infeasible. Representativeness is documented in 3.1.

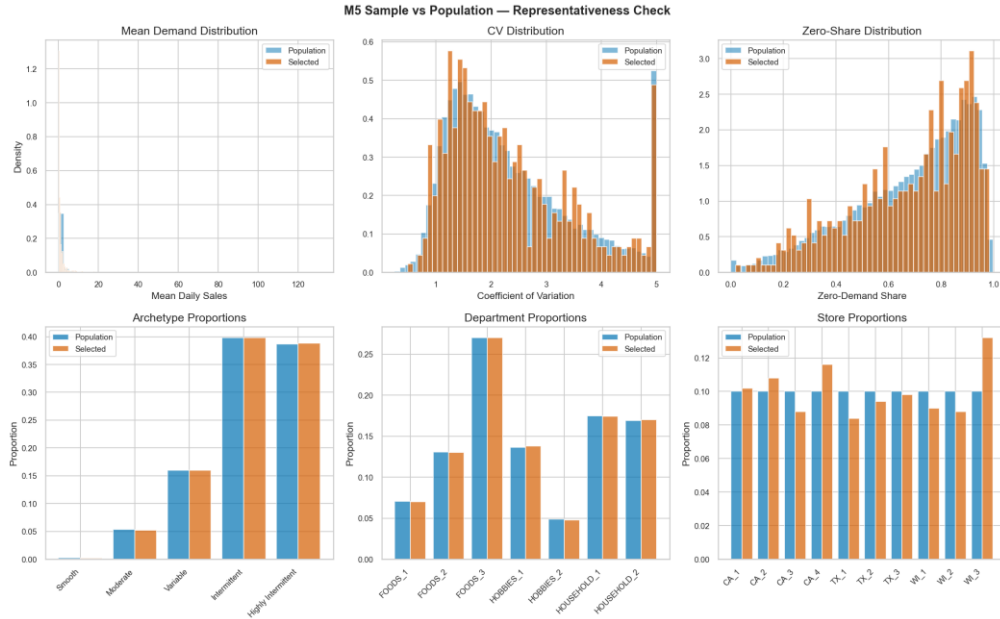


Figure 3.1: M5 500-series stratified selection: coverage across archetypes and departments. Source: notebook 04.

Data Preparation

M5 source files (`sales_train_evaluation.csv`, `calendar.csv`) were audited for dimensions (30,490 item-store series; day columns `d_1` onward with verified date-index alignment), and the Store file (`train.csv`) was verified as a complete panel: every store-item-date combination present, no duplicates, a single zero-sales entry. Both datasets were reindexed onto one common calendar so that cross-dataset comparison does not confound model effects with differences in history length.

Demand Archetypes

Following the Syntetos–Boylan tradition (Syntetos and Boylan 2005), and specifically its demand-pattern categorization (Syntetos et al. 2005), intermittency is characterized per series by average demand interval (ADI) and squared coefficient of variation (CV^2). The resulting quadrants are labeled here as Smooth, Intermittent, Erratic/Variable, and Lumpy/Highly Intermittent, with an additional Moderate band; volume terciles provide a second axis. M5 archetype counts are 194 highly intermittent, 199 intermittent, 26 moderate, 80 variable, and 1 smooth. The single smooth series is reported for completeness but supports no inference (section 6.4).

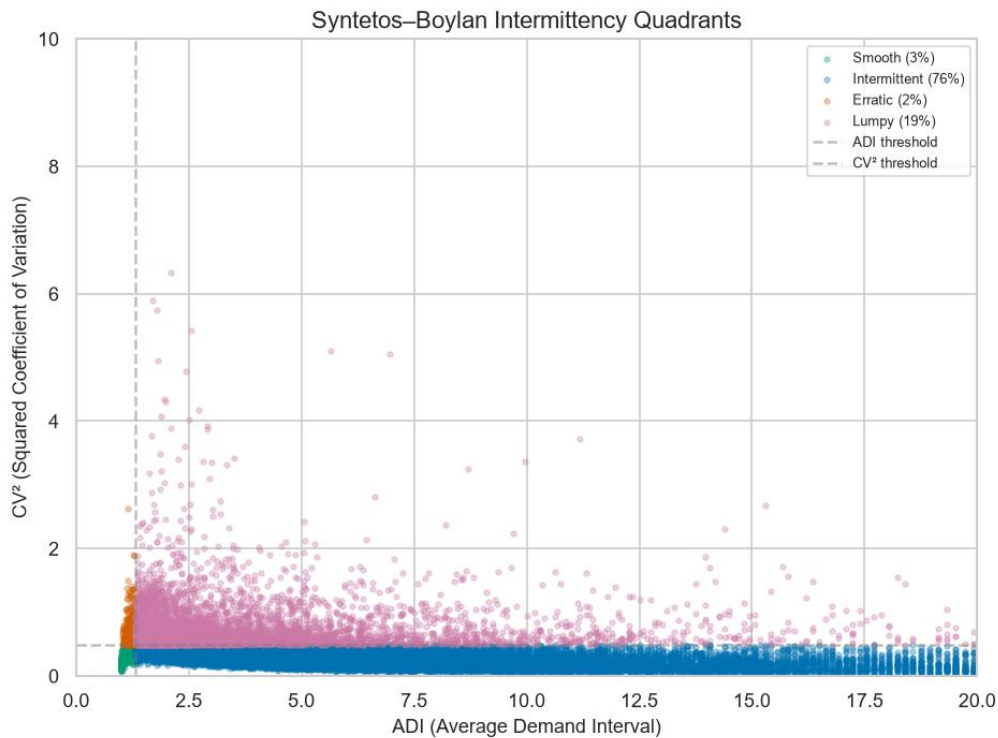


Figure 3.2: Intermittency quadrants (ADI versus CV), which motivate the archetype analysis and the Croston rung. Source: notebook 03.

Forecasting Models

Naive carries the last observed day forward 28 days. Seasonal Naive repeats the last seven observed days (an early implementation that reached beyond the seasonal window was identified and corrected to use observed values only). Moving Average uses validation-selected global windows (M5: 14 days; Store: 7 days; validation MAE 0.9206 and 7.4057 respectively). SES tracks a moving level ($\alpha = 0.1$), DES adds trend ($\alpha = \beta = 0.2$), and additive TES adds a weekly seasonal component (0.3/0.3/0.3); all use transparent implementations with validation-grid selection. ARIMA(1,1,0) runs on both datasets with a sparse-data pre-check (series with more than 85% zeros fall back to Naive rather than fitting noise); per-series order search was rejected as overfitting-prone and computationally out of scope. SARIMA(1,1,0)(0,1,1)₇ runs Store-only, since seasonal differencing of predominantly zero series lacks methodological justification and the Croston family already covers the M5 specialty. Croston, SBA, and TSB run M5-only (validation parameters 0.1, 0.5, and 0.1–0.3), as flat intermittent-demand forecasts have no applicability to dense data. The LSTM is one global network per dataset with a 28-day lookback, per-series history-only scaling, and chronological per-series validation (rebuilt during hardening to remove overlapping-window leakage), seeded at 42. Full configurations are in Annexure B.

Models are ordered by complexity while evaluation is held constant: baselines (Naive, Seasonal Naive, Moving Average) define the threshold any method must clear; SES, DES (Holt), and additive TES add memory for level, trend, and seasonality; ARIMA adds autocorrelation modeling and SARIMA seasonal autocorrelation; the Croston family specializes in intermittent demand; the global LSTM learns nonlinear patterns pooled across all series of a dataset. Complexity is the independent variable; whether it produces value is the empirical question. 3.3 summarizes the arrangement.

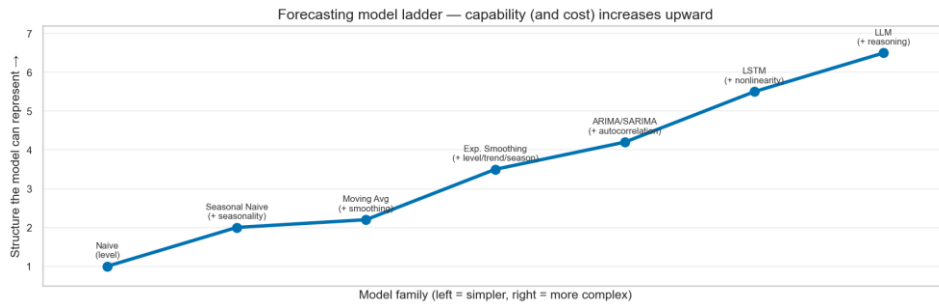


Figure 3.3: The model ladder: complexity increases by rung. Whether value increases with it is tested in section 4. Teaching figure from notebook 05a.

Each subsection states what the method assumes, how it was configured in this study, and what comparison it supports. Textbook derivations are omitted; implementation truth is in Annexure B.

Naive

Naive repeats the last observed value across the 28-day horizon, using only history preceding each origin. It is the reference floor: a method that cannot improve on repeating yesterday adds no value. Its accuracy is modest (M5 MAE 1.18; Store 16.29), yet it wins 6 of 27 Store inventory policies, an early indication that accuracy and cost can diverge.

Seasonal Naive

Seasonal Naive repeats the same weekday from the preceding week, the cheapest use of retail's weekly rhythm (figure 3.5). The implementation repeats only the last seven observed values; an earlier version that reached beyond the seasonal window was identified and corrected during hardening. It serves as the seasonal reference run in every phase. Despite middling accuracy, it is the third-lowest-cost Store inventory input (2,128.14), which suggests weekly structure stocks shelves effectively even when point accuracy is unremarkable.

Moving Average

The moving average forecasts the mean of the last w days, a low-pass filter on noisy demand and the standard practitioner baseline: no estimated dynamics to break. The global window was selected on validation (M5: $w = 14$ at MAE 0.9206, against 0.9236 for $w = 7$ and 0.9282 for $w = 28$; Store: $w = 7$ at 7.4057, against 7.5113 and 7.9132), with seed 42. It is the simple method the remaining rungs must justify themselves against. It nearly leads M5 forecasting (MASE 1.346 against LSTM's 1.316) and records the lowest Store inventory cost (2,084.50).

Simple Exponential Smoothing

SES tracks a slowly moving demand level with higher weight on recent days (Hyndman and Athanasopoulos 2018); it is the minimal model with memory, estimating level without trend or seasonality. The transparent implementation uses validation-selected $\alpha = 0.1$. SES performs consistently well without leading any forecast board: second-lowest cost on M5 inventory (159.31) and on Store inventory (2,113.82), and winner of 2 M5 sensitivity policies at short lead with high stockout penalty.

Double Exponential Smoothing

DES (Holt's method) extends SES with a trend component ($\alpha = \beta = 0.2$, grid-selected), testing whether explicit trend modeling helps. It does not help here. DES carries roughly double the inventory of the leading methods (average stock 8.41 on M5 and 76.07 on Store) while recording mid-range or worse costs, indicating that trend extrapolation on noisy retail data accumulates stock without commensurate accuracy. Its 5 Store sensitivity wins reflect Store leadership fragmentation rather than DES strength.

Triple Exponential Smoothing

TES (additive Holt-Winters, 0.3/0.3/0.3) adds a weekly seasonal component to DES, completing the classical toolkit in one model. It records the weakest results in the study: M5 MASE 2.60 with cost 346.68, Store MASE 2.11 with cost 3,419.88, and the highest error in every archetype. The finding is that combined trend-seasonality structure, as implemented here, overfits the available signal.

ARIMA

ARIMA models the next value as a linear combination of recent values and recent shocks after differencing toward stationarity (Box and Jenkins 1970; Hyndman and Athanasopoulos 2018), providing the statistical treatment of autocorrelation that smoothing lacks. A fixed ARIMA(1,1,0) was adopted after a validation sanity check; per-series order search was rejected as overfitting-prone and computationally out of scope. Series with more than 85% zeros skip estimation and fall back to Naive rather than fitting noise. ARIMA is the linear-memory representative: mid-range accuracy on both datasets (M5 MASE 1.51; Store 1.45) with a marked feasibility asymmetry discussed in section 5.1 (22.65% pre-check fallback on sparse M5 demand against zero fallbacks on dense Store demand).

SARIMA

SARIMA adds seasonal autoregression and moving-average terms, testing whether explicit weekly autocorrelation improves on SES-style seasonality for dense data. The configuration is SARIMA(1,1,0)(0,1,1)₇, Store-only, estimated on the full 500 series by 8 origins (4,000 estimations, 4,000 successful, 2,382 seconds total). A prior 100-series subset is retained as an exploratory record but is not the headline result. SARIMA was not estimated on M5: seasonal differencing of predominantly zero series lacks justification, and the Croston rung already addresses that setting. It is the second-ranked Store forecaster (MASE 1.055) but ranks fifth on Store inventory cost (2,282.31).

LSTM

The LSTM is a recurrent network with gating that carries signal across long, noisy sequences (Hochreiter and Schmidhuber 1997). Estimated globally (one network per dataset, 28-day lookback, per-series history-only scaling, chronological per-series validation rebuilt during hardening to remove overlapping-window leakage, seed 42), it tests whether pooled cross-series learning outperforms per-series statistics. Within the evaluated model set it records the lowest forecast error on both datasets and the lowest M5 inventory cost, but ranks fourth on Store inventory cost; section 5.1 examines this split in detail.



Figure 3.4: The LSTM cell: gates control what is retained, discarded, and emitted at each step, allowing information to persist across weeks of noisy demand. Teaching figure from notebook 08.

Croston

Croston’s method (Croston 1972) separates intermittent demand into two sequences (nonzero demand size and interval between nonzero demands) and smooths each independently, avoiding fitting zeros as signal. Estimated M5-only with validation-selected $\alpha = 0.1$; the 28-day forecast is flat at the estimated rate. Accuracy is respectable (MAE 1.08) and its M5 service level is the highest recorded (93.9%), but its cost (174.81) exceeds its bias-corrected variants below.

SBA

The Syntetos-Boylan approximation (Syntetos and Boylan 2005) multiplies the Croston rate by $(1 - \alpha/2)$ to correct its over-forecast bias. Estimated M5-only with $\alpha = 0.5$. It is the lowest-cost member of the family (160.66): a small algebraic correction with measurable inventory value.

TSB

The Teunter-Syntetos-Babai method (Teunter et al. 2011) replaces Croston’s interval smoothing with direct updating of demand-occurrence probability, responding faster when demand patterns shift. Estimated M5-only at $(\alpha, \beta) = (0.1, 0.3)$. It records the family’s lowest forecast error (MAE 1.010, MASE 1.369) but the family’s second-lowest cost (165.13): accuracy and cost rank the family differently.

Model-Ladder Comparison

3.1 compresses the ladder: twelve methods on five rungs across two datasets, with the two deliberate scope exceptions (SARIMA Store-only, Croston family M5-only). The ladder orders complexity; whether value follows is resolved empirically in section 4.

Table 3.1: Model taxonomy: purpose, scope, and comparative role. Full configurations in Annexure B.

Model	Rung	Scope	Comparative role
Naive	Baseline	Both	Reference floor
Seasonal Naive	Baseline	Both	Weekly-rhythm reference
Moving Average	Baseline	Both	Practitioner benchmark
SES	Smoothing	Both	Level memory
DES	Smoothing	Both	Trend specification (records overstocking)
TES	Smoothing	Both	Full classical specification (weakest results)
ARIMA(1,1,0)	Statistical	Both	Linear memory
SARIMA(1,1,0)(0,1,1) ₇	Statistical	Store only	Seasonal challenger

Model	Rung	Scope	Comparative role
LSTM (global)	Neural	Both	Pooled nonlinear learning
Croston	Intermittent	M5 only	Sparse-data specialist
SBA	Intermittent	M5 only	Debiased specialist
TSB	Intermittent	M5 only	Adaptive specialist

Experimental Design

3.2 states the frozen design. Splits are time-based only. Training covers 2013-01-01 to 2015-10-31 (1,034 days); validation 2015-11-01 to 2016-02-29 (121 days, leap day included after the hardening correction); testing 2016-03-01 to 2016-05-22 (83 days). Forecasts span $h = 28$ days from each of 8 weekly rolling origins under an expanding window, with each model restricted strictly to history preceding each origin (asserted in code). All model selection (averaging windows, smoothing weights, ARIMA orders, Croston parameters, LSTM stopping) uses validation data only; the test window is used once for final scoring. The global seed is 42 throughout.

Table 3.2: Frozen experimental protocol (source: 05_experiments/config.json).

Element	Frozen value
Common window	2013-01-01 to 2016-05-22 (1,238 days)
Train / validation / test	1,034 / 121 / 83 days (validation includes 2016-02-29)
Horizon h	28 days, 8 weekly rolling origins, expanding window
M5 sample	500 series, stratified by archetype $\times$ department
Store sample	All 500 series (complete panel)
Evaluation population	$500 \times 8 \times 28 = 112,000$ forecasts per model-dataset
Seed	42 throughout

Forecast Evaluation Metrics

Headline accuracy is mean absolute error (MAE); the primary scale-free metric is MASE with seasonal period $m = 7$ (the weekly retail rhythm shown in figure 3.5), whose denominator is the in-sample one-step seasonal-naïve error computed on training data only. Validation and test data never enter the denominator. RMSE, RMSSE, WAPE, and sMAPE are reported alongside; no metric is selected after seeing results. Two conventions matter for interpretation. First, zero denominators (constant training history) are left as honest missing values rather than epsilon-adjusted (no such cases occurred: $n_{\text{NaN}} = 0$ throughout). Second, archetype-level WAPE and sMAPE are pooled ratios over each cell, because averaging per-series ratios allows a single near-zero-demand series to dominate the mean; model-level WAPE columns remain per-series means and were verified consistent across files. sMAPE is reported with its known distortions on near-zero demand documented. Definitions are formalized in Annexure C.

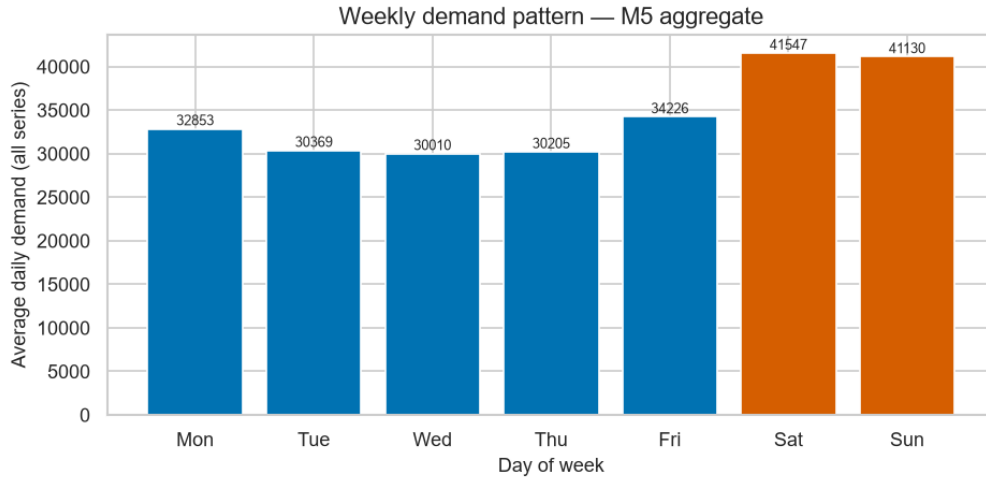


Figure 3.5: Weekly seasonality in retail demand, supporting seasonal period in MASE/RMSSE, Seasonal Naive, and SARIMA. Source: notebook 03.

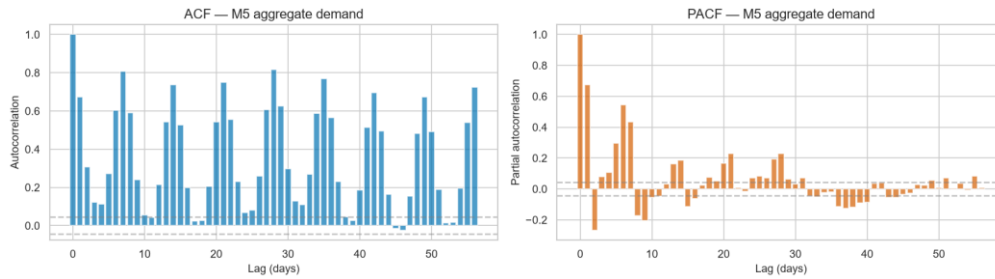


Figure 3.6: Autocorrelation structure: demand covaries with its recent past, which is the property ARIMA-style models exploit. Source: notebook 03.

Statistical Testing

Ninety-one paired comparisons (55 M5 and 36 Store) combine the Wilcoxon signed-rank test (normal approximation) with the Diebold-Mariano test (Diebold and Mariano 1995), Holm correction (Holm 1979) for multiplicity, 500-replication bootstrap confidence intervals (Efron and Tibshirani 1993), paired Cohen d_z , and the Wilcoxon rank-biserial r . With 112,000 paired errors, tests have high power to detect small differences, so each p -value is reported together with an effect size. A qualification on the observation count is required: the 112,000 paired errors per comparison arise from 500 series observed across 8 rolling origins and 28 horizon steps, with overlapping estimation windows. They are repeated, dependent observations, not 112,000 independent draws, and the nominal count should not be read as an effective independent sample size. The paired tests were preregistered on this structure and remain the reported inference; series-level or block-bootstrap inference that respects the dependence explicitly is identified as a robustness extension in section 6.6. Methodological detail is in Annexure C.

Inventory Policy

All forecasts pass through one shared ordering implementation (`11_src/inventory_policy.py`), identical for every model by construction: a daily-review order-up-to (base-stock) policy with lost sales. For each series-origin instance, safety stock equals z times the forecast-error standard deviation (floored) times $\sqrt{L}$; the order-up-to level is the sum of the next L forecasts plus safety stock;

unmet demand is lost at penalty P per unit; end-of-day stock accrues holding cost H per unit. The default ranking policy is $L = 7$, 95% service target ($z \approx 1.645$), $H = 1$, $P = 5$. Total cost decomposes into holding and stockout components, with service level, average inventory, and reorder frequency as diagnostics. 3.7 illustrates how forecasts enter ordering decisions.

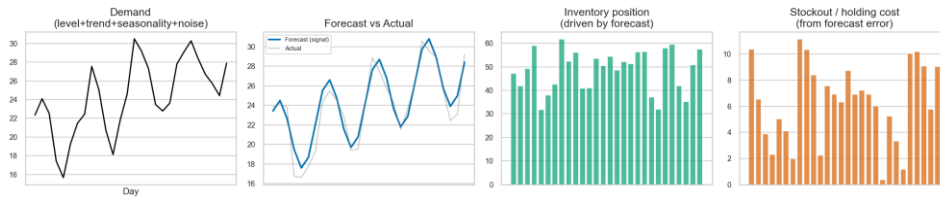


Figure 3.7: From forecast to order: forecasts determine safety stock and order-up-to levels, so the policy governs what each unit of error costs. Teaching figure from notebook 05a.

Sensitivity Analysis

Headline rankings use the default policy. Their stability is then tested on a 27-policy grid ($L \in \{3,7,14\}$, service $\in \{90\%, 95\%, 99\%\}$, $P \in \{3,5,10\}$, $H = 1$), replaying every model-dataset through 540 costing runs. A model ranked first across nearly all policies is described as robust; one whose rank moves with the policy is described as policy-sensitive. Per-archetype M5 costing adds the demand-structure dimension.

Validation and Reliability

Every ARIMA/SARIMA estimation is logged per series-origin (convergence_details_*.csv), with a scope column separating the exploratory 100-series SARIMA subset from the primary full-500 run. A 16-fit feasibility probe (0.62 s per fit) preceded the 4,000-fit full-500 SARIMA run (2,382 s; 4,000 of 4,000 successful). The contrast between complete convergence on dense data and partial pre-check fallback on sparse data is itself an empirical result about estimation feasibility under demand structure.

Reproducibility and Data Provenance

Before reporting, the evidence passed a hardening audit that corrected the calendar (leap day 2016-02-29 assigned to validation, giving 1,034/121/83), revalidated the LSTM on chronological per-series splits, fixed a Seasonal Naive lookahead, unified all models onto one inventory module, corrected MASE grouping and Wilcoxon effect sizes, pooled archetype WAPE/sMAPE, replaced the SARIMA subset with the full-500 run as the primary result, and repaired identifiers. Pre-hardening originals are preserved untouched in 06_results/_pre_hardening_backup/ and are not report sources. The freeze (commits 53c8343 through edb42ad) generated the number sheet, figure and table audits, and claim traceability programmatically; no headline number in this report was entered by hand. Hardening is documented here to establish the provenance of the evidence.

Data Analysis and Interpretation

Exploratory Data Analysis

Before any model comparison, both environments were characterized. M5 demand is sparse and intermittent, with a 64.5% zero rate across the studied series, and its archetype composition (194 highly intermittent, 199 intermittent, 26 moderate, 80 variable, 1 smooth series) motivates both the Croston rung and the archetype-level analysis of Section 5.1.0.1. Store demand is dense (0.02% zeros) with a pronounced weekly rhythm, wide level and volatility differences across series, and autocorrelation structure that ARIMA-style models exploit. The representative-series figures below, the intermittency quadrants (3.2), and the weekly-seasonality and autocorrelation panels ([fig:weekly,fig:acf]) fix this setting; all are descriptive, and evaluation begins in section 4.4.

M5 Demand Analysis

M5 (primary). Walmart M5 competition data (Makridakis et al. 2022): thousands of item-store daily series, 2011 to 2016. This study uses 500 stratified series over the common window 2013-01-01 to 2016-05-22. The archetype split is 194 highly intermittent, 199 intermittent, 26 moderate, 80 variable, and 1 smooth series. 4.1 illustrates the demand character.

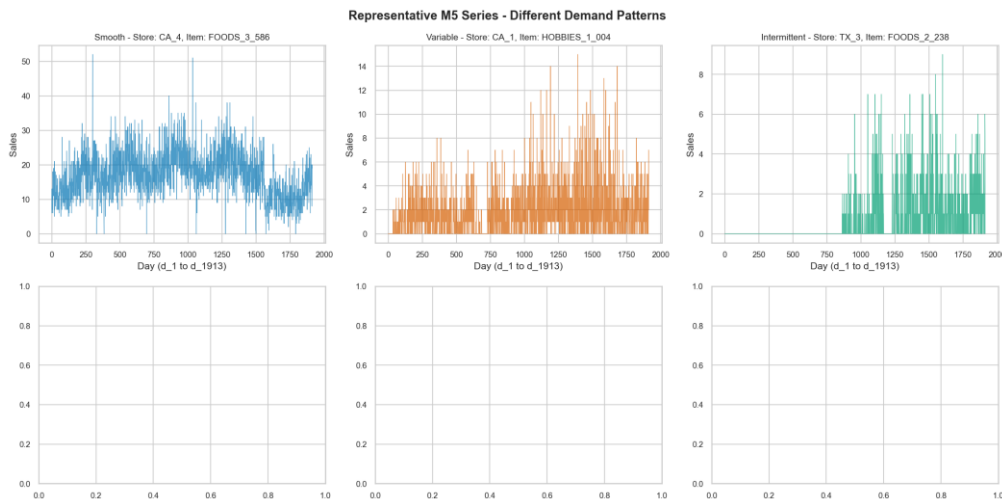


Figure 4.1: Representative sparse M5 series: extended zero-demand intervals punctuated by spikes. Source: 07_figures/eda/m5 (notebook 03).

Store Demand Analysis

Store Item Demand (secondary robustness environment). Kaggle Store Item Demand Forecasting Challenge: 10 stores by 50 items (500 series), a complete panel of 913,000 rows from 2013-01-01 to 2017-12-31, analyzed over the same common window. Demand is dense, with essentially no zero-demand days. [fig:rep-store,fig:diversity] illustrate the contrast.

Dataset 02 — Six representative series

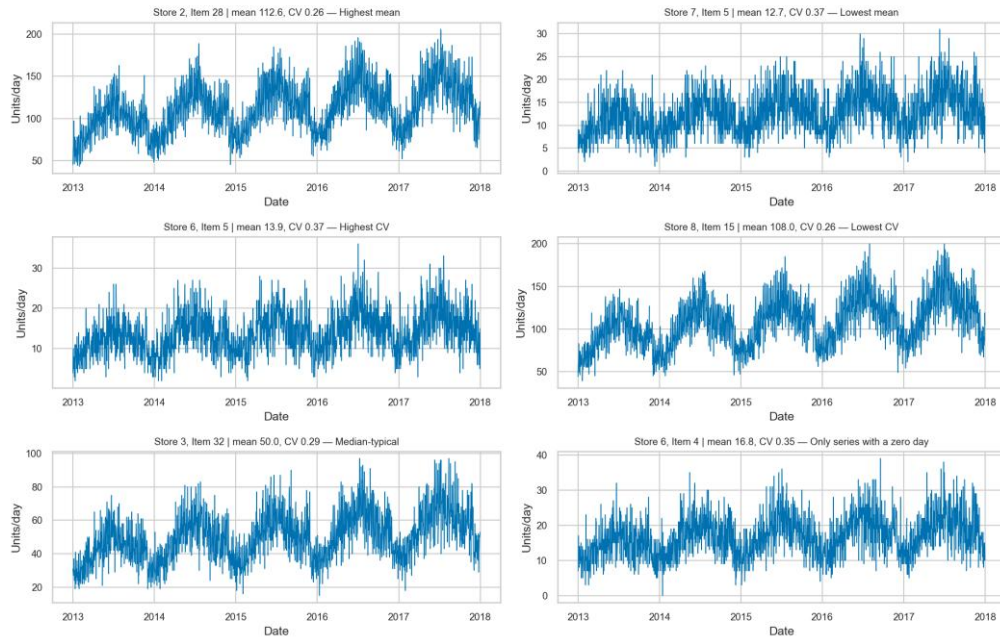


Figure 4.2: Representative dense Store series: nonzero demand each day with a weekly rhythm. Source: 07_figures/eda/store_item_demand (notebook 03).

Dataset 02 — 20 series sampled across the mean-demand spectrum

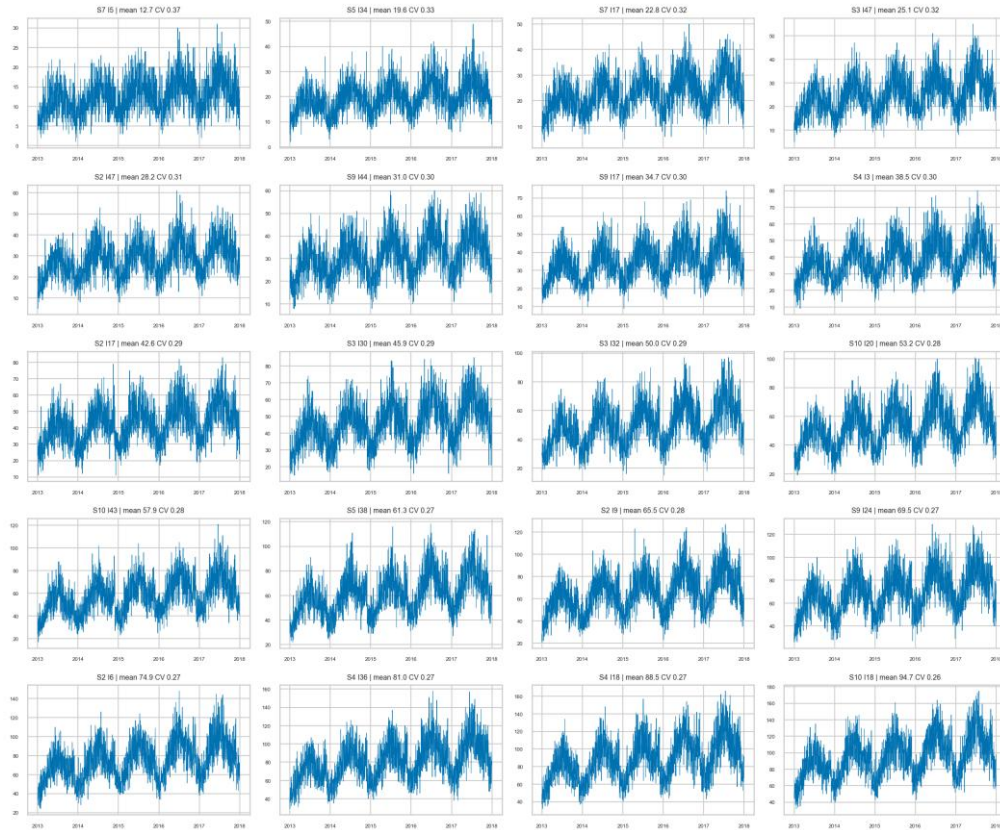


Figure 4.3: Store demand diversity: level and volatility differ widely across series within a dense environment. Source: notebook 03.

Forecasting Results

4.4 summarizes forecasting performance on one scale. On M5, a lead cluster (LSTM at 1.316, Moving Average at 1.346, SES at 1.351, TSB at 1.369) separates from ARIMA (1.505), the Naive pair (approximately 1.53), and the smoothing failures DES (1.987) and TES (2.599). On Store demand the spread is wider: LSTM (0.978) is the only model below 1.0, followed by SARIMA-full500 (1.055), Seasonal Naive (1.183), and Moving Average (1.191), with Naive (1.971) and TES (2.111) at the upper end. MASE below 1.0 indicates improvement over the training-period seasonal-naive benchmark, which only Store LSTM achieves. The tight M5 cluster requires formal testing to rank internally (section 4.7.1).

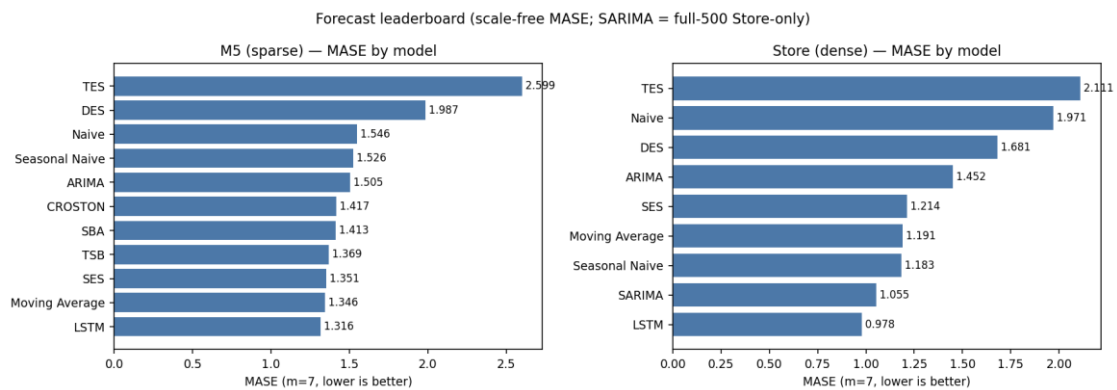


Figure 4.4: Forecast leaderboard (MASE, lower is better). M5 shows a close LSTM, Moving Average, SES, and TSB cluster; Store shows LSTM below 1.0 with SARIMA-full500 second. Deterministic rebuild from final files (F-NEW-01).

M5 Forecast Results

4.1 reports the M5 board. LSTM leads on MAE (0.989) and MASE (1.316); Moving Average (0.9997 / 1.346) and SES (0.998 / 1.351) are close behind, with TSB (1.010 / 1.369) completing the lead group. SBA and Croston follow; ARIMA (1.132 / 1.505) trails without reaching Naive-pair levels; Naive (1.184) and Seasonal Naive (1.187) anchor the middle; DES (1.514) and TES (1.922) record the highest errors. The RMSE column adds information: Moving Average records the lowest RMSE by a wide margin (1.30), since averaging suppresses the large errors RMSE penalizes quadratically, yet it does not lead on MAE or MASE. Metric choice already affects rankings before inventory enters the analysis. sMAPE (approximately 117–161%) confirms that percentage errors on near-zero demand are large for all methods; per-series mean WAPE ranks LSTM (0.77) first, consistent with MAE.

Table 4.1: M5 forecast leaderboard (per model). Bold: column best.

Model	MAE	RMSE	MASE	RMSSE	WAPE	sMAPE (%)
LSTM	0.99	1.95	1.32	0.85	0.77	144.2
Moving Average	1.00	1.30	1.35	0.86	1.31	143.5
SES	1.00	2.00	1.35	0.86	0.78	145.1
TSB	1.01	2.01	1.37	0.88	0.79	146.3
SBA	1.07	2.12	1.41	0.89	0.83	148.1
Croston	1.08	2.11	1.42	0.87	0.85	143.9
ARIMA	1.13	2.32	1.51	1.05	0.88	139.0

Model	MAE	RMSE	MASE	RMSSE	WAPE	sMAPE (%)
Naive	1.18	1.60	1.55	1.12	1.43	160.9
Seasonal Naive	1.19	1.67	1.53	1.14	1.45	152.7
DES	1.51	3.43	1.99	1.33	1.18	117.8
TES	1.92	4.39	2.60	1.85	1.50	116.0

Note: the M5 ARIMA row is a hybrid evaluation combining 3,094 fitted ARIMA forecasts with 906 Naive fallback forecasts triggered by the documented greater-than-85%-zero-demand pre-check (see section 5.1).

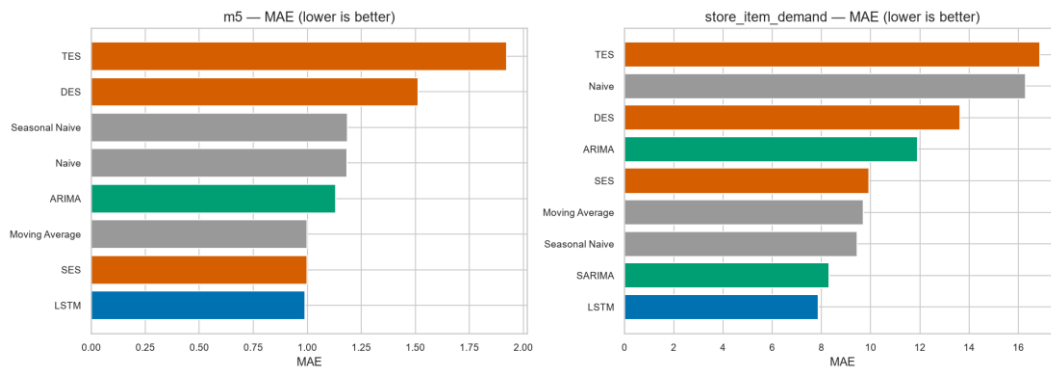


Figure 4.5: Full-ladder MAE comparison including LSTM (hardened run). The M5 cluster near 1.0 and Store’s LSTM-led spread are both visible; exact boards are tables 4.1 and 4.2. Source: notebook 08.

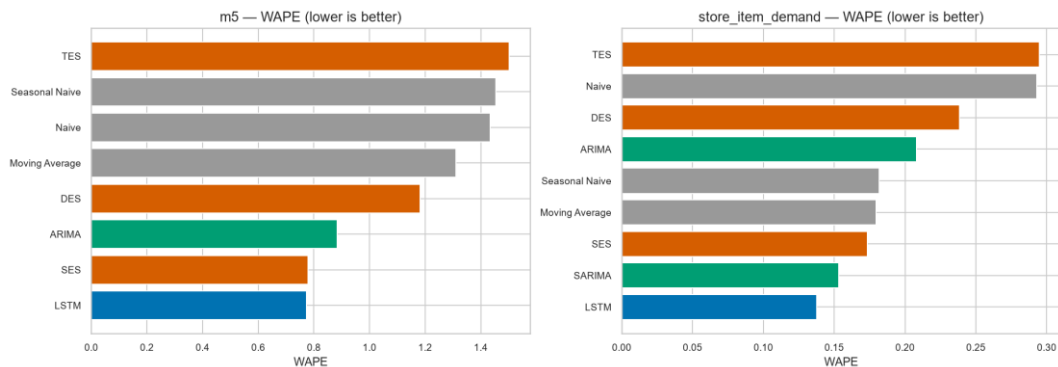


Figure 4.6: Full-ladder WAPE comparison. Store WAPE levels are an order of magnitude smaller (dense demand raises the denominator), so cross-dataset WAPE levels are not comparable; ranks are. Source: notebook 08.

Store Forecast Results

4.2 reports the Store board, where ordering is less ambiguous. LSTM leads on MAE (7.87), MASE (0.978), RMSSE (0.971), and WAPE (0.138); SARIMA-full500 is second (8.45 / 1.055 / 1.040 / 0.148); Seasonal Naive (9.45) and Moving Average (9.71) form the next tier, followed by SES (9.92); ARIMA (11.90), DES (13.63), Naive (16.29), and TES (16.87) trail. Two observations matter. First, seasonal structure is informative on Store demand: SARIMA’s seasonal moving-average term reduces MAE from ARIMA’s 11.90 to 8.45, and Seasonal Naive outperforms SES. Second, full500 SARIMA (MAE 8.45, MASE 1.055) differs modestly from the exploratory 100-series subset (8.31 and approximately 1.069): a population effect, documented rather than adjusted, which is why subset results do not headline.

Table 4.2: Store forecast leaderboard (per model; SARIMA full500). Bold: column best.

Model	MAE	RMSE	MASE	RMSSE	WAPE	sMAPE (%)
LSTM	7.87	10.42	0.98	0.97	0.14	15.3
SARIMA	8.45	11.09	1.05	1.04	0.15	16.9
Seasonal Naive	9.45	11.64	1.18	1.17	0.18	19.2
Moving Average	9.71	12.08	1.19	1.18	0.18	18.3
SES	9.92	13.41	1.21	1.21	0.17	18.6
ARIMA	11.90	15.98	1.45	1.43	0.21	23.0
DES	13.63	18.44	1.68	1.67	0.24	25.4
Naive	16.29	18.95	1.97	1.87	0.29	33.2
TES	16.87	23.39	2.11	2.16	0.29	33.7

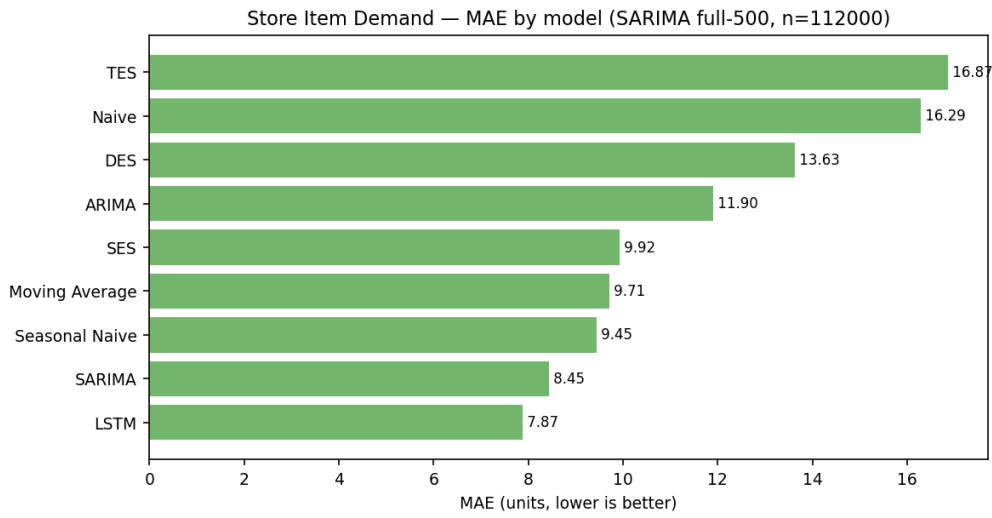


Figure 4.7: Store MAE with full-population SARIMA (8.45,), replacing the subset-based panels. Deterministic rebuild (F-NEW-02).

Statistical Comparisons

With 112,000 paired errors per comparison, tests have high power to detect small differences, so table 4.3 reports effect sizes alongside p -values. Two rows illustrate why. M5 LSTM versus TSB: $d_z \approx -0.031$ with Holm-adjusted $p \approx 2 \times 10^{-10}$. The difference is significant in the formal sense and negligible in absolute terms (two hundredths of an MAE unit). Store LSTM versus Moving Average: $d_z \approx -0.29$ with $p \approx 0$, a difference that is both significant and material at nearly two MAE units. The same significance label covers a tenfold difference in magnitude. A p -value addresses whether models differ; d_z , the rank-biserial r , and the bootstrap interval address whether the difference matters. The full 91-pair table is Annexure F; formal methodology is in Annexure C.

Table 4.3: Spotlight pairwise comparisons (paired-error = A minus B; negative favors A).

Dataset	A vs. B	d_z	Holm p	r	95% CI
M5	LSTM vs. TSB	−0.031	2.0×10^{-10}	−0.020	[−0.024, −0.017]
M5	LSTM vs. MA	−0.015	1.8×10^{-3}	−0.010	[−0.014, −0.007]
M5	LSTM vs. SES	−0.014	5.1×10^{-4}	−0.011	[−0.013, −0.005]

Dataset	A vs. B	d_z	Holm p	r	95% CI
M5	MA vs. SES	+0.006	1.7×10^{-12}	-0.022	[0.000, 0.003]
Store	LSTM vs. MA	-0.295	≈ 0	-0.282	[-1.868, -1.800]
Store	LSTM vs. SARIMA	-0.118	≈ 0	-0.128	[-0.599, -0.548]
Store	LSTM vs. SES	-0.336	≈ 0	-0.311	[-2.085, -2.014]

*Wilcoxon significant but Diebold-Mariano $p = 0.056$ (raw) and 0.113 (Holm): the tests disagree on MA versus SES, and the pair is reported as indistinguishable for practical purposes.

Pairwise statistical comparisons

Three patterns hold across all 91 pairs. First, the M5 lead cluster is genuinely ordered (LSTM ahead of Moving Average, SES, and TSB, each pairwise significant) but the gaps are operationally negligible; the cluster spans 0.05 MASE. Second, Store gaps are an order of magnitude larger, and every LSTM pairing is significant and material. Third, MA versus SES on M5 is the study's reference null case: significant by Wilcoxon, not significant by Diebold-Mariano (Holm $p = 0.113$), with an interval spanning zero. Two strong simple methods with nothing substantive between them, recorded as such even at $n = 112,000$.

Inventory Results

The shared engine from section 3.11 supplies the decision treatment: every forecast drives the same daily-review order-up-to policy ($L = 7$, 95% target, $H = 1$, $P = 5$) with lost sales, where total cost equals holding plus stockout cost. Rankings below are total cost under that policy; service level, average inventory, and reorder frequency record how each cost was composed. Sensitivity of rankings to the policy itself is tested separately in sections 4.13.1 and 4.13.

M5 Inventory Results

4.4 ranks M5 decision cost. LSTM is lowest (152.83), followed by Moving Average (158.83) and SES (159.31); SBA (160.66), TSB (165.13), and Croston (174.81) follow; Seasonal Naive (192.55), ARIMA (194.60), and Naive (200.75) form the middle group; DES (268.80) and TES (346.67) record substantially higher costs. Cost composition varies instructively: SBA's low holding cost (128.39) is offset by stockouts (32.27); SES records the second-lowest stockout component (21.00, after Croston at 20.28) with the third-highest service level (93.6%), after Croston and Seasonal Naive; Croston records the highest service level (93.9%) at intermediate cost. LSTM's cost margin over Moving Average (approximately 6 units) indicates that the inventory policy can amplify relatively small forecast differences into larger cost differences when demand is sparse.

Table 4.4: M5 inventory ranking under the shared policy (cost units; lower better). Bold: best.

Model	Total	Holding	Stockout	Service (%)	Avg. inv.
LSTM	152.83	130.04	22.80	92.8	4.64
Moving Average	158.83	136.30	22.52	93.0	4.87
SES	159.31	138.32	21.00	93.6	4.94
SBA	160.66	128.39	32.27	90.9	4.59

Model	Total	Holding	Stockout	Service (%)	Avg. inv.
TSB	165.13	143.54	21.59	93.2	5.13
Croston	174.81	154.54	20.28	93.9	5.52
Seasonal Naive	192.55	170.73	21.82	93.8	6.10
ARIMA	194.60	162.62	31.98	89.6	5.81
Naive	200.75	158.25	42.50	87.2	5.65
DES	268.80	235.35	33.46	89.9	8.41
TES	346.67	304.28	42.40	88.0	10.87

Store Inventory Results

4.5 ranks Store decision cost and differs from the forecast board. Moving Average is lowest (2,084.50); SES (2,113.83) and Seasonal Naive (2,128.15) complete the lowest three with no LSTM among them; LSTM (2,247.46) ranks fourth, SARIMA (2,282.31) fifth, ARIMA (2,320.87) sixth; Naive (2,700.37), DES (2,779.64), and TES (3,419.88) follow. Cost composition reverses the M5 pattern: Naive holds the least stock (28.38) but incurs the highest stockout component (1,905.74) at 43.2% service; DES and TES carry high inventory (76.07 and 88.77) while still incurring mid-range stockouts. Moving Average is comparatively balanced: moderate stock (45.58), a low stockout frequency (10.72, fourth-lowest), and no extreme in either column. This ranking holds under the default policy and is conditional on it; section 4.13.1 tests its stability and must be read alongside any citation of this table.

Table 4.5: Store inventory ranking under the shared policy. Bold: best. Stability across policies is tested in section 4.13.1.

Model	Total	Holding	Stockout	Service (%)	Avg. inv.
Moving Average	2,084.50	1,276.10	808.40	61.7	45.58
SES	2,113.83	1,155.60	958.24	56.1	41.27
Seasonal Naive	2,128.15	1,281.05	847.10	62.0	45.75
LSTM	2,247.46	1,157.70	1,089.76	53.1	41.35
SARIMA	2,282.31	1,307.78	974.53	59.2	46.71
ARIMA	2,320.87	1,076.48	1,244.40	52.4	38.45
Naive	2,700.37	794.63	1,905.74	43.2	28.38
DES	2,779.64	2,130.09	649.55	73.5	76.07
TES	3,419.88	2,485.65	934.23	69.1	88.77

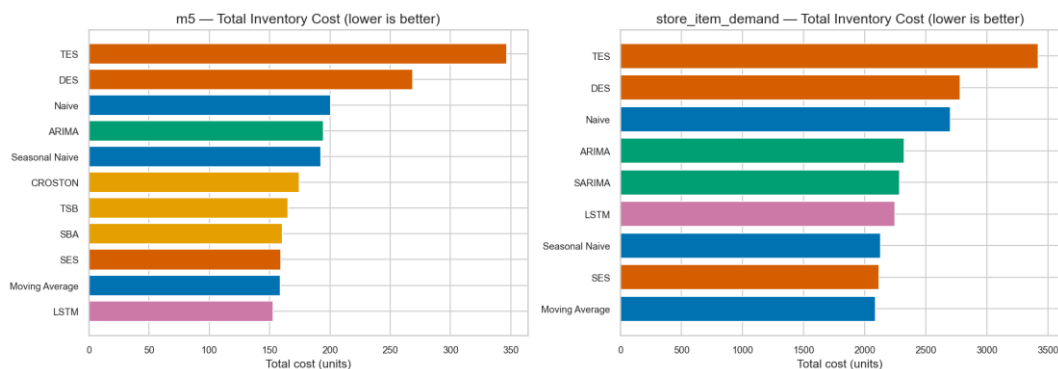


Figure 4.8: Total inventory cost by model and dataset under the shared policy. M5: LSTM lowest. Store: Moving Average lowest, LSTM fourth. Source: unified simulation (notebook inventory layer).

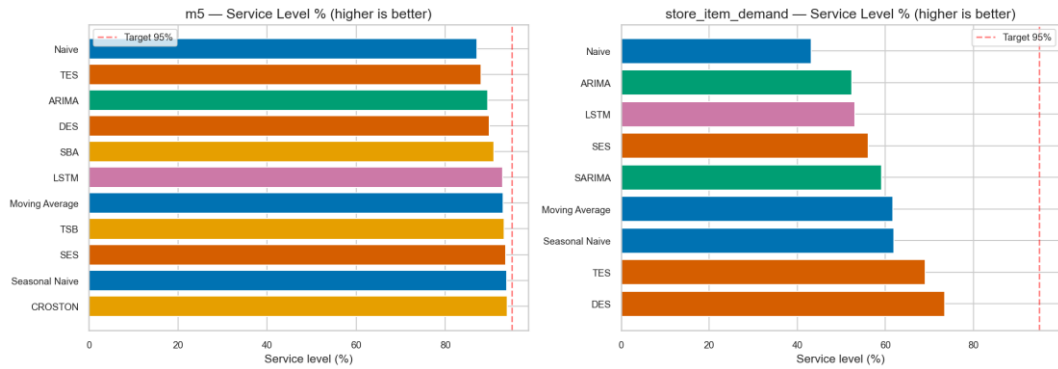


Figure 4.9: Service level by model: Croston highest on M5 (93.9%); all models record lower service on Store demand (43–74%), reflecting denser, higher-volume exposure. Same source as figure 4.8.

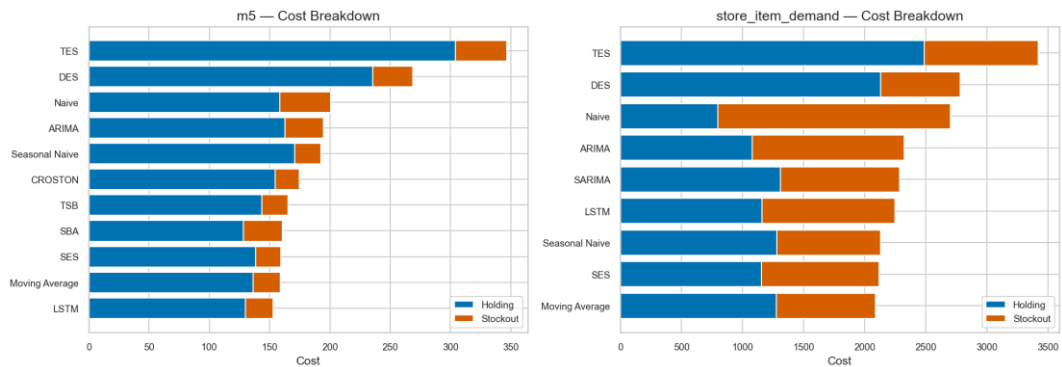


Figure 4.10: Holding versus stockout composition. Naive's Store cost is almost entirely stockout; DES and TES costs are holding-heavy on both datasets. Same source as figure 4.8.

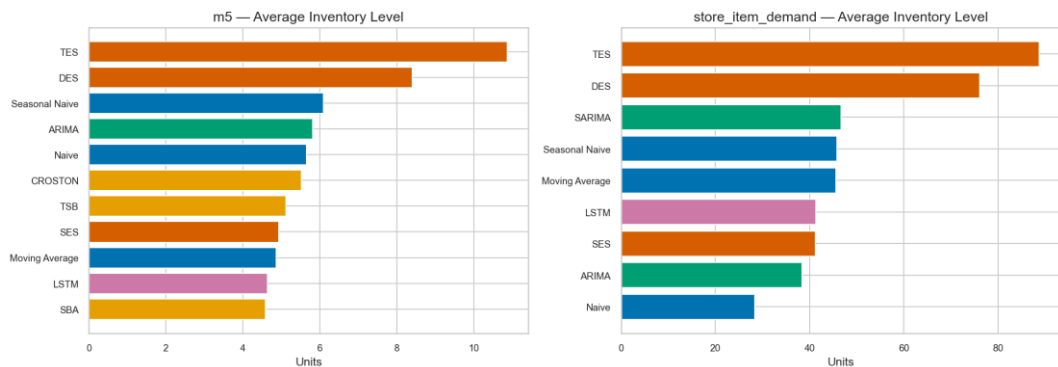


Figure 4.11: Average inventory and reorder frequency: DES and TES carry roughly double the stock of the leading methods. Same source as figure 4.8.

Forecast Accuracy vs. Inventory Cost

4.12 plots forecast error against inventory cost per model. On M5 the association runs in the expected direction: lower error accompanies lower cost, with LSTM at the favorable corner. On Store demand it does not: LSTM is leftmost (lowest error) but mid-ranked vertically, while Moving Average, SES, and Seasonal Naive sit to its right and below it. The mechanism is cost asymmetry. The policy charges under-prediction (lost sales at $P = 5$) five times the rate of over-prediction (holding at $H = 1$), and safety stock scales with each model's own error volatility. A forecaster that averages toward the middle and rarely under-shoots can therefore record lower cost than a sharper forecaster with occasional shortfalls. Accuracy is a symmetric criterion; the inventory objective is not.

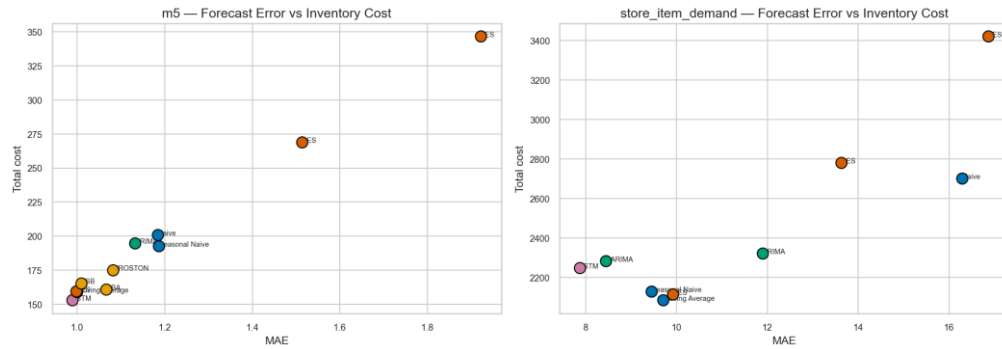


Figure 4.12: Forecast error versus inventory cost per model. M5: lower error accompanies lower cost. Store: LSTM records the lowest error but mid-ranking cost. Same source as figure 4.8.

Croston-Family Analysis

The specialists support their inclusion. On M5, TSB records the family's lowest forecast error (MAE 1.010, MASE 1.369; fourth overall), followed by SBA (1.066) and Croston (1.082): adaptivity ranks ahead of bias correction, which ranks ahead of the original, on accuracy. On cost the order differs: SBA (160.66), TSB (165.13), Croston (174.81). SBA's debiasing trims the over-forecast that holding cost penalizes, while TSB's responsiveness carries somewhat higher stock. Within a single family, then, accuracy and cost already rank methods differently. Substantively, the family validates the design choice: on intermittent demand, methods that decline to fit zeros belong in the comparison.

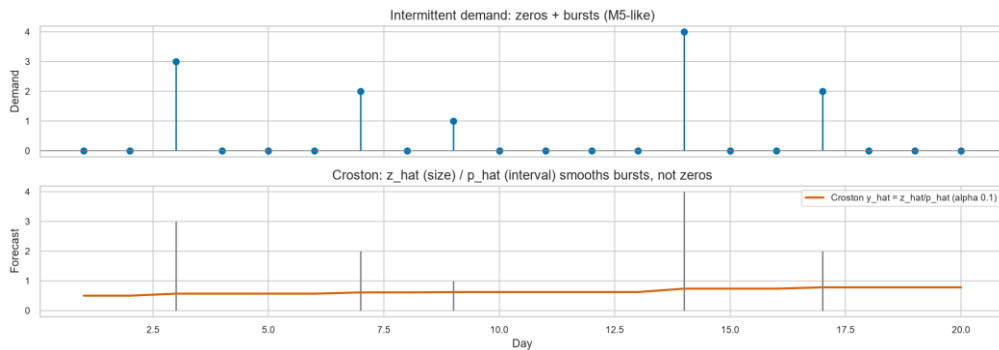


Figure 4.13: Croston intuition: sparse demand is separated into nonzero demand size and interval between nonzero demands, each smoothed independently. Teaching figure from notebook 06c.

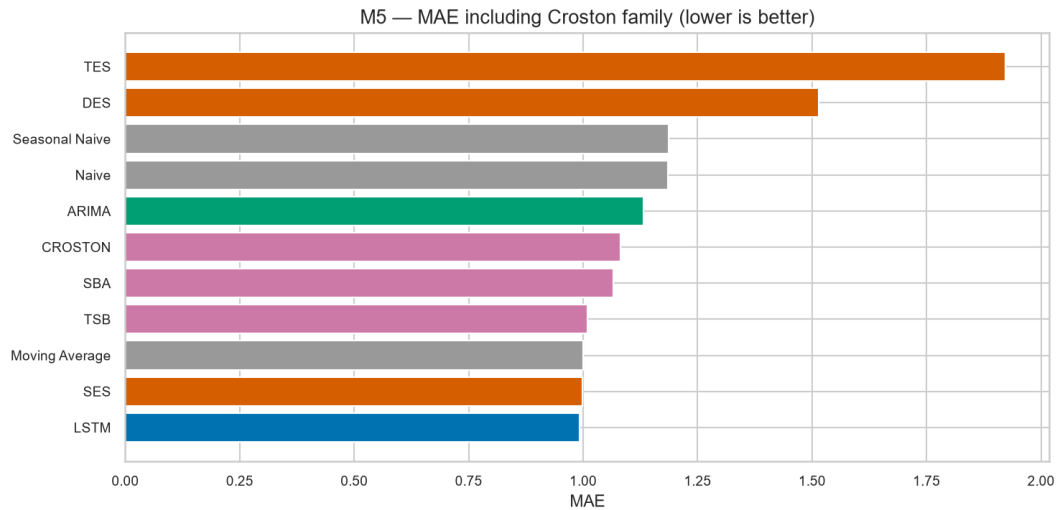


Figure 4.14: M5 MAE with the Croston family: TSB lowest in the family at 1.010. Source: notebook 06c.

Sensitivity Analysis

Robustness across inventory policies

4.6 with figures 4.15 and 4.16 summarizes stability. M5: LSTM ranks first in 25 of 27 policies. Its two defeats, both to SES, occur in one region: short lead time ($L = 3$) with high stockout penalty ($P = 10$), where SES's conservative level-tracking incurs less emergency cover. Store: Moving Average 9, SES 7, Naive 6, DES 5. No model reaches half the grid; the lowest-cost model is a function of the policy, not a property of any model. M5 conclusions therefore support deployment under stated conditions; Store conclusions support policy-conditional selection.

Table 4.6: Robustness summary: policy wins across the 27-policy grid. M5 stable; Store policy-sensitive.

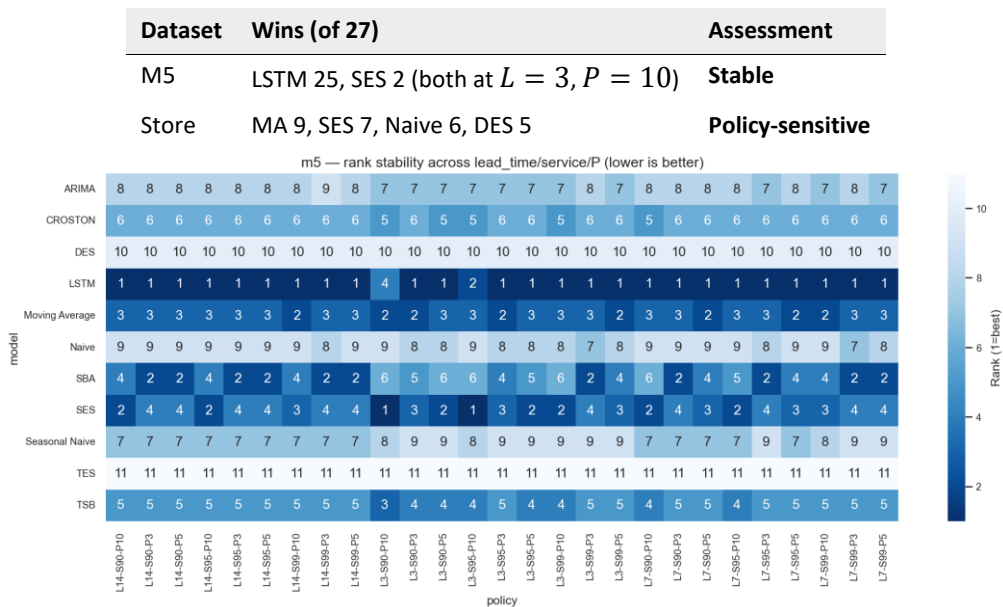


Figure 4.15: M5 rank heatmap across 27 policies: LSTM first throughout except two SES cells in the short-lead, high-penalty corner. Source: sensitivity layer.

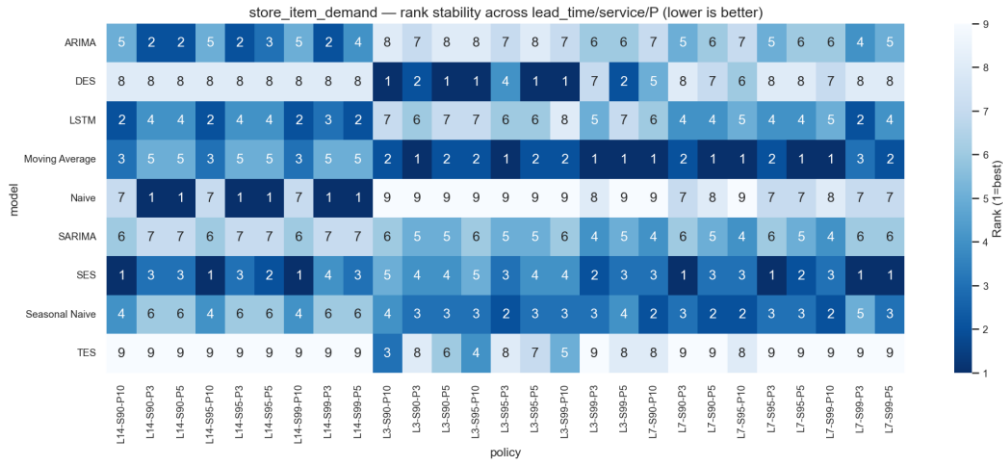


Figure 4.16: Store rank heatmap: lowest cost moves across four models as the policy varies. Same source as figure 4.15.

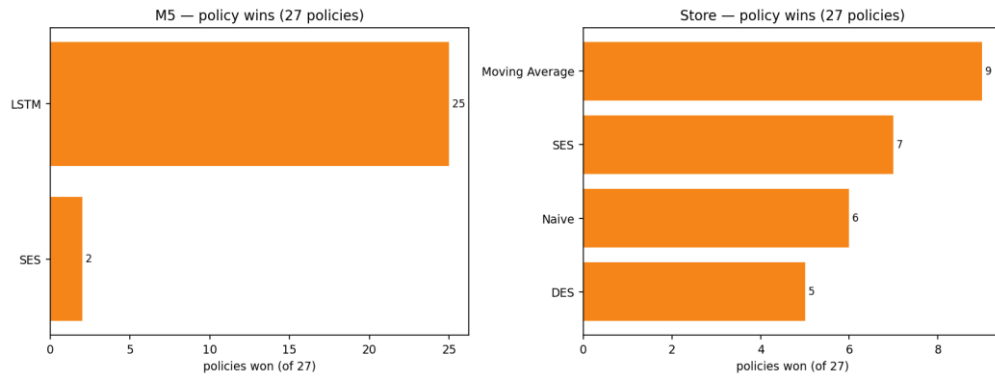


Figure 4.17: Winner shares across the 27-policy grid: M5 concentration against Store fragmentation. Deterministic rebuild (F-NEW-03).

The grid interior indicates where leadership changes. Longer lead times amplify safety-stock differences, disadvantaging volatility-sensitive models; higher P/H penalizes under-predictors (Naive deteriorates except in short-lead Store cells where its leanness helps); higher service targets favor conservative levels (SES gains). Per-archetype M5 costing (55 rows, `sensitivity_by_archetype_m5.csv`) confirms the structural pattern: sparse archetypes magnify each policy effect. Full matrices are Annexure G.

Convergence and Feasibility

4.7 accounts for every estimation. M5 ARIMA: 3,094 fits, 906 pre-check deferrals (22.65%), 0 optimizer failures. Store ARIMA: 4,000 of 4,000. SARIMA subset (exploratory): 800 of 800. SARIMA full500 (primary): 4,000 of 4,000 in 2,382 seconds. The feasibility gradient (dense settings estimate throughout; sparse settings estimate selectively) describes the data, not the software.

Table 4.7: Convergence accounting. Pre-check deferrals are specified guards, not failures.

Scope	Model	Attempted	Failed	Naive fallback	Fallback %
M5 full500	ARIMA	4,000	0	906	22.65
Store full500	ARIMA	4,000	0	0	0.00
Store subset (explor.)	SARIMA	800	0	0	0.00
Store full500 (primary)	SARIMA	4,000	0	0	0.00

Interpretation of Results

Read together, the boards of this chapter tell one story. Forecast accuracy favours the neural rung on both datasets; inventory cost under the shared policy agrees on M5 (LSTM lowest, 152.83) and disagrees on Store (Moving Average lowest, 2,084.50; LSTM fourth, 2,247.46). The mechanism is cost asymmetry: accuracy is a symmetric criterion, while the policy charges under-prediction five times the rate of over-prediction and scales safety stock with each model's own error volatility (section 4.11). Statistical testing confirms the orderings are real while cautioning that significance is not magnitude (section 4.7.1), and the 27-policy grid shows the Store disagreement is structural: leadership there is fragmented across four models (9/7/6/5), so the lowest-cost input is a function of the policy, not a property of any model (section 4.13.1). The findings, their interpretation, and the recommendations they license are consolidated in Chapter 5.

Findings and Recommendations

Key Findings

Accuracy metrics are symmetric, per-unit, and policy-free; inventory costs are asymmetric, thresholded, and policy-defined. Safety stock converts error volatility into money; the order-up-to rule converts error bias into money; and the P/H ratio sets the exchange rate. An evaluation that stops at the forecast therefore measures only part of a model's economic effect.

Demand-Archetype Findings

[fig:arch-mae,fig:arch-wape] disaggregate M5 accuracy by demand shape using pooled WAPE/sMAPE, since per-series ratios would let near-zero series dominate the mean. LSTM records the lowest MAE in Highly Intermittent (0.416) and Intermittent (0.989) demand. In Variable demand the top three (SES at 1.799, Moving Average at 1.804, LSTM at 1.806) lie within 0.007 of each other, a difference without practical content. SES records the lowest MAE in Moderate demand (2.634). TES records the highest error in every archetype, including the smoother shapes its trend-seasonality specification might be expected to fit. The Smooth column rests on a single series (LSTM nominally lowest at forecast MAE 2.358) and is shown for completeness without supporting conclusions. Inventory cost by archetype (figure 5.4) distributes the lowest cost across methods: LSTM in Highly Intermittent (64.52) and Intermittent (147.94), Moving Average in Variable (285.93), and SBA in Moderate (383.59). No single method records the lowest cost in every demand shape.

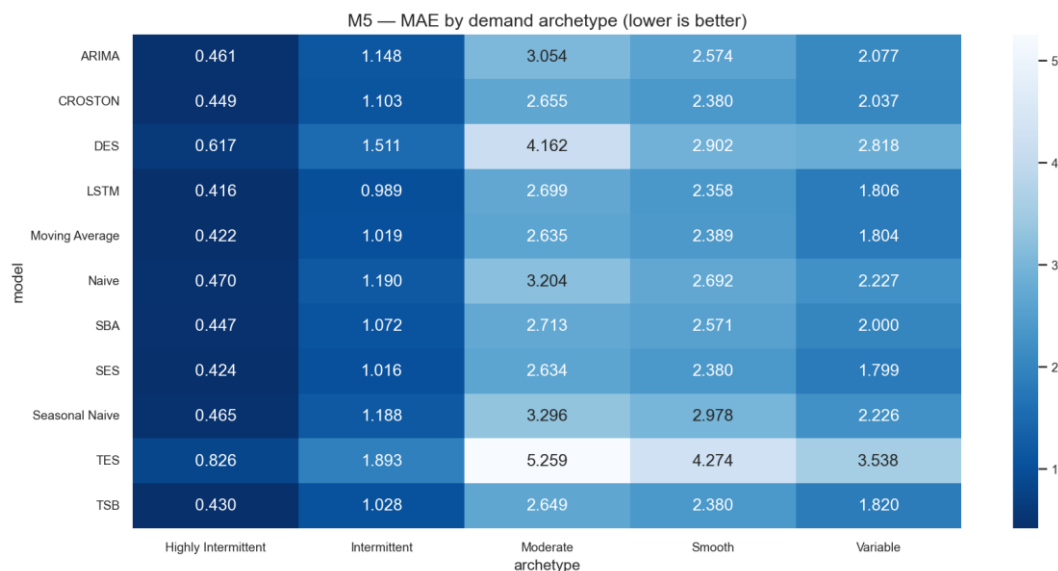


Figure 5.1: M5 MAE by archetype: LSTM lowest on sparse shapes; Variable is effectively tied; TES highest throughout. Pooled-metric figure (notebook archetype layer).

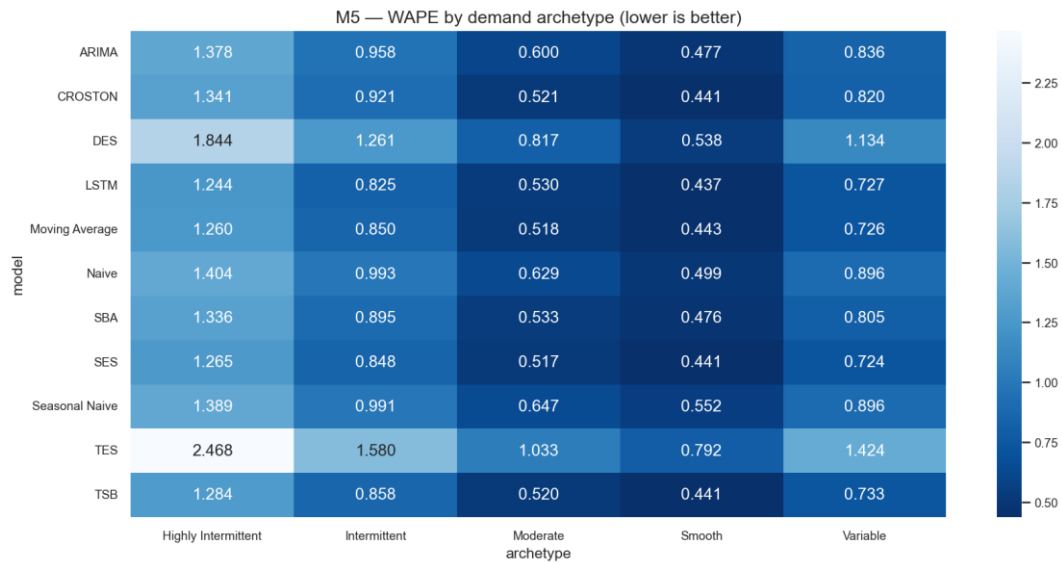


Figure 5.2: M5 pooled WAPE by archetype: the ratio-of-sums construction prevents near-zero series from dominating the mean. Same source as figure 5.1.

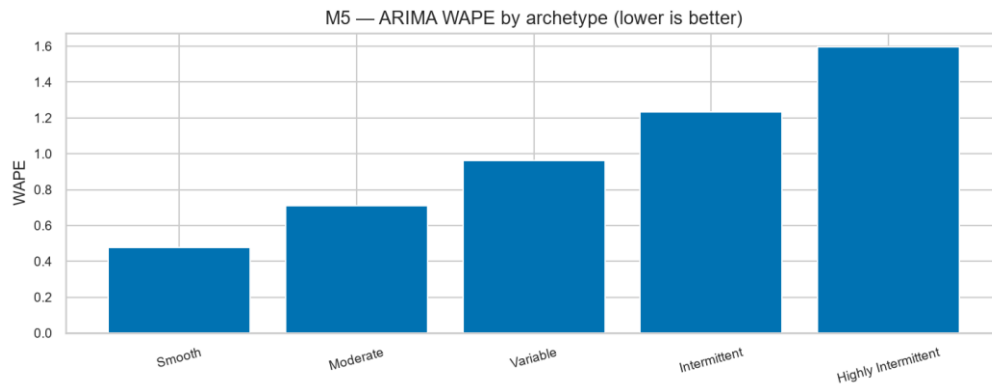


Figure 5.3: ARIMA error by M5 archetype: error increases where demand is sparsest, consistent with the 22.65% fallback rate. Source: notebook 07.

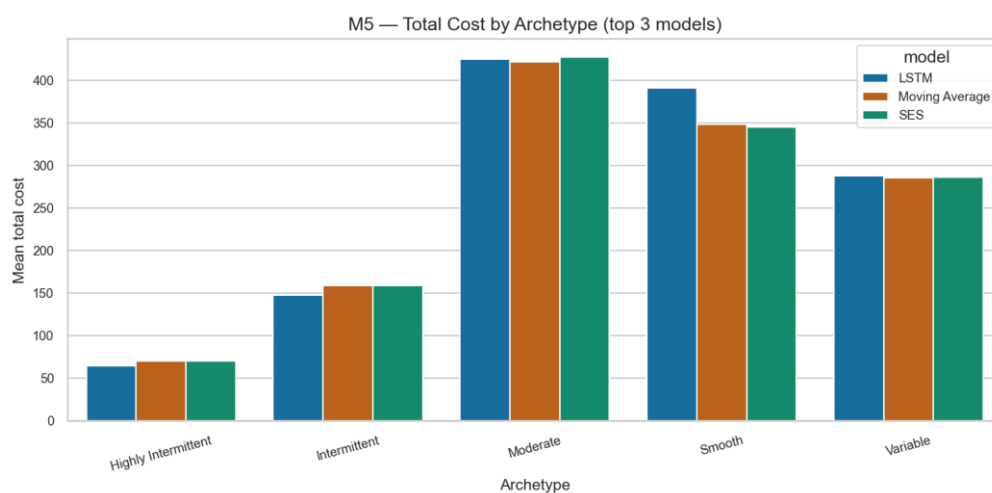


Figure 5.4: M5 inventory cost by archetype: lowest cost by method and shape (LSTM on sparse shapes; Moving Average on Variable; SBA on Moderate). Same source as figure 4.8.

LSTM Findings

The neural rung partly confirms its premise. As a forecaster, the global LSTM records the lowest MAE and MASE on both datasets, the lowest archetype MAE on sparse shapes, and the only sub-1.0 MASE on Store demand. As an inventory input it is split: lowest cost on M5 (152.83, first in 25 of 27 policies) but fourth on Store (2,247.46), where its sharper forecasts include under-predictions that the $P/H = 5$ cost ratio penalizes heavily. On provenance: the LSTM was revalidated on chronological per-series splits after overlapping-window leakage was identified in the first validation design, so these figures reflect generalization under the corrected protocol rather than validation leakage. For practice, the implication is conditional: forecast quality is established across both environments, while inventory value should be assessed against the applicable cost ratio.

ARIMA Findings

ARIMA occupies an unstable middle position. On dense Store data it estimates cleanly (4,000 of 4,000 fits, zero fallbacks) yet ranks sixth on both boards (MASE 1.452; cost 2,320.87), indicating that linear memory contributes little where weekly structure dominates. On sparse M5 data, 906 of 4,000 estimations (22.65%) trigger the greater-than-85%-zero pre-check and defer to Naive, concentrated in Highly Intermittent series (figure 5.3); there are zero optimizer failures, so every non-skipped estimation converges. The pre-check is a specified guardrail rather than an estimation failure, and its asymmetry (22.65% against 0%) measures how demand structure conditions statistical feasibility.

SARIMA Full-Population Findings

The seasonal challenger performs as specified. All 4,000 Store estimations succeed with zero fallbacks in 2,382 seconds (approximately 0.6 seconds per fit): seasonal differencing at $m = 7$ is feasible at full population, not only on the exploratory 100-series subset (MAE 8.31) that preceded it. Final full-500 accuracy (MAE 8.45, MASE 1.055, RMSSE 1.040) differs slightly from the subset, a population effect recorded without adjustment. Decision quality (2,282.31, fifth) lags accuracy (second): improved weekly tracking still under-protects against stockouts at $P/H = 5$. SARIMA was not estimated on M5, deliberately: seasonal differencing of predominantly zero series lacks methodological justification, and the Croston rung already addresses that setting.

Model Complexity Findings

H3 in monotone form is rejected. The most complex classical specifications record the weakest results (TES last on all boards; DES an overstocking liability), the mid-complexity linear model sits mid-table, and only the neural model converts complexity into accuracy, and then into inventory value on one of two datasets. The simplest rung (Moving Average, SES) wins or nearly wins three of the four boards. Complexity purchases capability (the LSTM's cross-series learning is measurable), but value is determined jointly with demand structure and cost asymmetry.

Negative and Surprising Findings

Null and negative results are reported explicitly. (1) TES ranks last on all boards. (2) DES and TES carry roughly double the inventory of the leading methods ($1.7\times$ and $2.2\times$, respectively) at lower service than SES. (3) Naive, the simplest benchmark, wins 6 Store policies. (4) Seasonal Naive is the third-lowest-cost Store inventory input. (5) The LSTM trails a moving average on Store inventory cost. (6) Archetype-level WAPE and sMAPE required pooled construction because per-series averaging is

unstable on near-zero-demand series; see section 3.9 for the correction and its rationale. (7) MA versus SES on M5 is indistinguishable for practical purposes despite $n = 112,000$ paired errors (subject to the dependence qualification in section 3.10).

Interpretation: Why LSTM Wins Forecasting but Not Always Inventory

The global LSTM captures patterns unavailable to per-series linear models: zero-inflated regimes, cross-item weekly structure, and nonlinear rebounds. That accounts for its forecast lead on both datasets. Inventory, however, purchases neither point forecasts nor patterns directly. It purchases a dispersion summary (error volatility, via safety stock) and a level (the next- L sum). On dense Store demand, the LSTM's sharpness includes occasional confident under-predictions: inexpensive under MAE, expensive at $P = 5$. The moving average, by construction, rarely under-shoots by large amounts. The relevant distinction is therefore between precision and caution under asymmetric cost, not between intelligence and simplicity.

Interpretation: Why Simple Models Remain Competitive

Moving Average and SES satisfy two conditions the policy rewards. First, averaging is close to optimal when most daily variation is unpredictable noise; the recent mean is then a minimum-variance guess. Second, their errors are approximately symmetric and stable, which is what safety-stock formulas assume. Simplicity aligns with the policy's assumptions. Sophistication improves on it only where the demand signal is strong enough to repay the departure, which holds on sparse M5 demand but not on dense Store demand under the default cost ratio.

Findings by Research Question

The research question asked how effectiveness changes as forecasting moves from traditional to neural methods when effectiveness is measured at inventory level. The evidence answers it in four movements. On accuracy, the neural rung leads both datasets (MASE 1.316 on M5, 0.978 on Store; section 4.4). On cost, the answer splits by environment: correspondence on sparse M5 demand (LSTM lowest, 152.83, first in 25 of 27 policies) and divergence on dense Store demand (Moving Average lowest, 2,084.50; LSTM fourth, 2,247.46; sections 4.9 and 4.10). On inference, the orderings are statistically real, but the M5 lead-cluster gaps are operationally negligible while the Store gaps are material (section 4.7.1). On stability, M5 leadership is robust across 27 policies and Store leadership is fragmented 9/7/6/5 (section 4.13.1). Effectiveness therefore improves with sophistication in forecasting and only conditionally in inventory terms.

Hypothesis Evaluation

H1 (accuracy differs): confirmed; 91 paired tests, nearly all significant under Holm control. **H2 (accuracy moves inventory): confirmed with qualification;** accuracy moves inventory cost but not monotonically and not in the same direction on both datasets. **H3 (monotone sophistication): rejected;** TES, DES, and ARIMA trail simpler methods, and only LSTM complexity pays, partially. **H4 (cross-dataset consistency): rejected in strong form;** forecasting leadership transfers (LSTM on both), decision leadership does not (LSTM against Moving Average), and policy-stability does not transfer at all (stable against fragmented). The qualified verdict is the contribution: which properties transfer (forecast skill) and which do not (decision superiority, policy-stability) are now mapped.

Empirical and Analytical Contribution

The study measures the forecast-decision relationship in both directions within one controlled design: correspondence on intermittent M5 demand and divergence on dense Store demand, with the Store podium (Moving Average, SES, Seasonal Naive ahead of LSTM) as the sharpest case. It records where sophistication pays (LSTM accuracy on both datasets; LSTM cost on sparse demand) and where it does not (TES and DES; ARIMA on dense demand).

Ninety-one Holm-corrected paired comparisons with effect sizes and bootstrap intervals, reported under the rule that p -values do not travel without d_z . The MA-SES reference null shows the procedure returning equivalence as well as difference.

Methodological Contribution

Three reusable instruments. First, a shared-policy evaluation harness that holds the inventory treatment constant across models: one module, every model, no per-model tuning. Second, a leakage-safe rolling protocol (history strictly before origin, validation-only selection, honest missing-value handling in scale-free metrics with pooled archetype ratios). Third, a stability discipline in which headline rankings are published together with their 27-policy record, separating stable conclusions from conditional ones.

At $n = 112,000$ paired errors, significance is inexpensive and consequently uninformative on its own. The M5 LSTM-TSB pair ($p \approx 2 \times 10^{-10}$, $d_z = -0.031$) is significant in the formal sense and negligible for any business purpose. Effect sizes, intervals, and the MA-SES test disagreement are what prevent overinterpretation. This qualification is strengthened, not weakened, by the dependence structure noted in section 3.10: correlated observations inflate nominal precision further, which is an additional reason to weight d_z over p .

Managerial / Business Implications

For intermittent catalogs, learned or specialist forecasting (LSTM, TSB/SBA) is supported, and the forecast advantage reaches the shelf. For dense, fast-moving catalogs, simple averaging methods should be piloted against the neural option under the applicable cost ratio before engineering investment; the lowest-cost input may already be in production. In both settings, leaderboards that stop at accuracy are insufficient evidence for procurement.

Intermittency determines which machinery is legitimate. Where zeros dominate, methods that decline to fit them (the Croston family, the Naive-fallback ARIMA guardrail) and learners that pool across series (the LSTM) perform best, while linear autoregression lacks estimable structure. Where demand is dense and weekly, seasonal specifications (SARIMA, Seasonal Naive) contribute and trend specifications (DES, TES) overfit noise into stock. The archetype tables constitute the suitability map; the Store-M5 contrast corroborates it.

The ladder's complexity order predicts neither the accuracy order nor the cost order. The only complexity that pays is the neural kind, and only in accuracy everywhere and in cost in one environment. The practical reading is budgetary: a moving average costs nothing to operate and nearly leads; an LSTM costs substantial engineering and leads conditionally.

Model Selection Recommendations

Selection is conditional on three answers: the demand structure (sparse or dense), the cost asymmetry (P/H), and the lead time. The report's tables cover the remainder: sparse demand with standard policies favors LSTM; dense demand under the default policy favors Moving Average with SES as the alternative; dense demand with short leads and high penalties requires a fresh comparison, since leadership is fluid there.

The policy parameters ($L, z, P/H$) function as the effective loss: forecasters are implicitly optimized for symmetric error but scored on asymmetric cost. The sensitivity grid makes this visible, with the lowest-cost model changing as P moves from 3 to 10. Model selection should therefore be conditional on the policy, and accuracy-only leaderboards should be read with that conditionality stated.

Stability is a property of conclusions under policy variation, not of models. M5-LSTM is stable (25 of 27, with losses confined to one explicable region). Store-MA is policy-sensitive (9 of 27), as is every Store alternative. Policy-sensitivity does not invalidate the default-policy result: under the stated policy, Moving Average genuinely records the lowest cost. It conditions the result: changing lead time or cost ratio requires re-running the comparison before acting on it.

Implementation Recommendations

Four implementation rules follow from the evidence. First, evaluate at inventory level: no procurement decision should be taken from an accuracy leaderboard alone, because accuracy metrics are symmetric while inventory costs are not (section 5.1). Second, match the method to demand structure: for intermittent catalogs, learned or specialist forecasting (LSTM, TSB/SBA) is supported; for dense, fast-moving catalogs, simple averaging methods should be piloted against the neural option under the applicable cost ratio before engineering investment (section 5.6). Third, re-test when the policy changes: leadership on dense demand is policy-sensitive, so changing lead times or cost ratios requires re-running the comparison before acting on it (section 5.7). Fourth, freeze and audit the evidence: the hardening lessons of section 6.3 show that the experiment requires the same scrutiny as the model.

Prioritised Recommendations

1. For sparse, intermittent assortments, deploy learned or specialist forecasting (LSTM on M5-style demand; SBA/TSB as lighter alternatives); the forecast advantage reaches cost and persists across policies.
2. For dense, smooth assortments, run the moving average (or SES) as the baseline-to-beat and require any neural candidate to beat it at inventory cost under the applicable P/H ratio, not on forecast error alone.
3. Treat every Store-style conclusion as policy-conditional: re-run the comparison when lead time, service target, or cost ratio changes.
4. Report effect sizes alongside every significance claim; at high-power sample sizes, significance alone does not establish practical importance.

5. Require inventory-level evaluation and a frozen, auditable evidence base for any model-selection decision.

Conclusion, Limitations and Future Scope

Overall Conclusion

Effectiveness does not track sophistication monotonically. Moving from traditional methods to neural forecasting improves forecast accuracy consistently (the LSTM records the lowest error on both datasets) but improves inventory outcomes only conditionally: substantially on sparse intermittent demand, and not at all on dense demand under the default cost structure. The model ladder purchases capability throughout and value in specific settings.

Answers to Research Questions

M5: LSTM lowest forecast error (MASE 1.316) and lowest inventory cost (152.83), ranking first in 25 of 27 policies. Store: LSTM lowest forecast error (MASE 0.978) but fourth-lowest inventory cost (2,247.46), behind Moving Average (2,084.50), SES, and Seasonal Naive; policy wins split 9/7/6/5. Ninety-one paired tests separate formal significance from magnitude (M5 LSTM-TSB: $p \approx 2 \times 10^{-10}$ at $d_z = -0.031$; Store LSTM-MA: $d_z = -0.29$), subject to the dependence qualification in section 3.10. H1 is confirmed, H2 confirmed with qualification, H3 rejected in monotone form, and H4 rejected in strong form.

Main Contributions

The study's contributions are empirical, methodological, and analytical (Sections 5.4 and 5.5); two further contributions concern reproducibility and research practice. The frozen package (number sheet, audits, traceability, validation log with 0 FAIL, pre-hardening backup, seeds, manifests) maps every reported number to a source file in one lookup. Provenance details are in Annexure I.

Research-Hardening Lessons

The costliest errors found were methodological rather than algorithmic: an unassigned leap day, overlapping validation windows, a subset result carried at full weight, and a mean-of-ratios construction unstable on near-zero series. Each was identified by audits that assumed nothing (row counts, date containment, denominator discipline, population labels). The general lesson is that the experiment requires the same scrutiny as the model, since conclusions are drawn from the former.

Limitations

The limitations below bound every conclusion in this report. They are stated explicitly so that the findings are read within, and not beyond, their evidence.

Dataset Scope

Two retail environments, however contrasted, remain two. Both reflect store-based daily demand; neither covers e-commerce spikes, perishables with expiry, promotion-driven regimes, or supply-side censoring beyond the lost-sales assumption.

Series Selection

M5 results describe the stratified 500, not the 30,490-series catalog. Stratification protects archetype coverage but does not guarantee catalog-level point estimates; extending the full ladder to the catalog is compute-bound (SARIMA alone required 2,382 seconds for 500 series).

Forecast Horizon

All conclusions are conditional on $h = 28$. Shorter horizons tend to favor naive and averaging methods; longer ones may favor trend and seasonal specifications. The robustness grid varies policy, not horizon.

Inventory-Policy Scope

One policy family (daily-review order-up-to, lost sales, no capacity limits, no minimum order quantities, no perishability, stationary costs) carries all decision conclusions. Backordering regimes, batch constraints, or stochastic lead times could reorder the boards. The stability discipline, rather than any single ranking, is the transferable element.

Feature Scope

Only univariate demand was used: no prices, promotions, calendar events, or cross-series covariates. Additional features could widen the LSTM's advantage or improve the classical methods with regressors. Both directions remain open questions.

Archetype Limitations

The Smooth archetype contains exactly one series ($n = 1$). Nothing in this report generalizes from it, and its figures appear for completeness only.

SARIMA Scope

SARIMA is Store-only by design; no claim is made about seasonal autoregression on intermittent demand. The 100-series subset is exploratory history, not the headline result.

Computational Considerations

Per-estimation costs (approximately 0.03 seconds for ARIMA and 0.6 seconds for SARIMA, plus LSTM training) bound full-catalog scaling. Per-series ARIMA order search was excluded on this ground; larger budgets could revisit it.

Statistical Dependence

As stated in section 3.10, the 112,000 paired errors per comparison are repeated observations from 500 series across overlapping origins and horizons, not independent draws. Reported significance levels therefore describe the preregistered paired design, not inference at a 112,000-strong independent sample size. This limitation applies symmetrically to all pairwise verdicts in the report.

What Can and Cannot Be Generalised

Transferable findings concern mechanisms and methods: the forecast-decision gap and its cost-asymmetry mechanism, the significance-versus-size discipline, the sensitivity-as-stability method,

LSTM forecast performance across both structures, and the TES/DES failure modes. Not transferable are the specific board winners (policy- and structure-contingent), anything about the Smooth archetype ($n = 1$), anything about LLMs (unexecuted), and anything beyond $h = 28$ or univariate demand. The contribution is a map with stated boundaries rather than a universal ranking.

Future Research

The LLM Experiment: Designed, Not Executed

The program's final rung (local LLM forecasting with controlled versus context-enhanced prompting, leak-guarded prompt construction in `11_src/llm_experiment_design.md`) was designed but deliberately not executed, held behind a frozen benchmark so the benchmark could not be adjusted against it. **No LLM result exists; none is presented, implied, or charted anywhere in this report.** Executing that phase against the frozen evidence base is the natural next step.

Proposed Future Extensions

In priority order: (1) execute the LLM phase; (2) add price, promotion, and calendar covariates; (3) widen the policy family (backorders, minimum order quantities, perishability); (4) vary the horizon (7 and 56 days); (5) extend the inexpensive rungs toward the full M5 catalog; (6) test additional retail environments; (7) replicate the pairwise inference with series-level and block-bootstrap procedures that model the dependence in section 6.4 explicitly.

Final Conclusion

For sparse, intermittent assortments, the learner is supported: the forecast advantage reaches cost and persists across policies. For dense, smooth assortments, a moving average should be tested against costlier options under the applicable cost ratio before investment, and any winner should be re-tested when lead times or penalties change. In all cases, procurement should require inventory-level evaluation rather than accepting an accuracy leaderboard.

Forecasting skill and decision value are related but distinct quantities, joined by demand structure and cost asymmetry. This study measured both, located where they correspond and where they part, stress-tested the results across 27 policies, and froze every number for audit. Whether the lowest-error forecaster is also the lowest-cost inventory input depends on the setting; the evidence now records the settings on each side.

Annexure A — Dataset Details

Machine truth: `05_experiments/config.json` (v1.0, 2026-08-28; validation end corrected 2016-02-28 → 2016-02-29 by recorded decision 2026-09-03). Common window 2013-01-01..2016-05-22 (1,238 d); train 2013-01-01..2015-10-31 (1,034); validation 2015-11-01..2016-02-29 (121); test 2016-03-01..2016-05-22 (83). Horizon 28, origin spacing 7, 8 origins (2016-03-01 through 2016-04-19), expanding window, history strictly before origin.

M5: 500 stratified series (`m5_series_selection.json`); Store: all 500. Forecast metrics MAE/RMSE/sMAPE/WAPE/MASE/RMSSE; inventory metrics holding/shortage/total cost, service level, average inventory, reorder count.

Governance: `00_project/decisions.md` (519 lines),
`00_project/research_readiness_checklist.md` (all PASS).

Annexure B — Model Configurations

Moving Average global windows: M5 $w = 14$ (validation MAE 0.9206 vs. 0.9236 for $w = 7$, 0.9282 for $w = 28$), Store $w = 7$ (7.4057 vs. 7.5113/7.9132); seed 42. SES $\alpha = 0.1$; DES $(\alpha, \beta) = (0.2, 0.2)$; additive TES (0.3, 0.3, 0.3); identical grids both datasets (`model_parameters.json`, `execution_metadata.json`). ARIMA(1,1,0) both datasets with $> 85\%$ -zero $\rightarrow$ Naive pre-check. SARIMA(1,1,0)(0,1,1)₇ Store-only, full500 (`run_log_full500.json`: 4,000 fits, 2,382 s, seed 42). Croston $\alpha = 0.1$, SBA $\alpha = 0.5$, TSB (0.1, 0.3) (`validation_selection.json`); validation window 2015-11-01..2016-02-29 (121 d) throughout. LSTM: global per dataset, lookback 28, history-only per-series scaling, chronological last- k -window validation, seed 42 (notebook 08). Inventory: shared module, POLICY_DEFAULT $L = 7$, 95%, $H = 1$, $P = 5$, σ floor 0.1; sensitivity grid $L \in \{3, 7, 14\} \times$ service $\{0.90, 0.95, 0.99\} \times P \in \{3, 5, 10\}$.

Annexure C — Statistical Output

For each model pair, paired errors $d_i = |e_i^A| - |e_i^B|$ ($n = 112,000$): Wilcoxon signed-rank (normal approximation) and Diebold–Mariano (Diebold and Mariano 1995) tests, Holm step-down correction (Holm 1979) across all 91 pairs, 500-replication bootstrap percentile CIs for mean $\bar{d}$, Cohen paired $d_z = \bar{d}/s_d$, and Wilcoxon rank-biserial r . “A better” in the raw file means $\bar{d} < 0$; the report’s verdicts additionally require material $|d_z|$ and interval support. Because nominal paired observations repeat across series, origins, and forecast horizons, Diebold–Mariano tests and bootstrap confidence intervals inherit this dependence structure; they are therefore interpreted alongside effect sizes rather than as evidence from independent observations. The MA–SES M5 disagreement (Wilcoxon significant, DM Holm $p = 0.113$) is the worked example of restraint.

Let y_t be actual demand, $\hat{y}_t$ the forecast, n the test size. $MAE = n^{-1} \sum |y_t - \hat{y}_t|$. With seasonal period $m = 7$ and training-only seasonal-naïve denominator $Q = (T - m)^{-1} \sum_{t=m+1}^T |y_t - y_{t-m}|$ over the frozen train window, $MASE = (n^{-1} \sum |y_t - \hat{y}_t|) / Q$ (Hyndman and Koehler 2006). RMSSE replaces absolute errors with squared ones under the root. WAPE (model level) is the mean of per-series ratios $\sum |y - \hat{y}| / \sum |y|$; WAPE (archetype level) is the pooled ratio $\sum_{\text{cell}} |y - \hat{y}| / \sum_{\text{cell}} |y|$, adopted after a mean-of-ratios computation exploded to $\sim 2 \times 10^9$ on near-zero-test-demand series. $sMAPE = 100 n^{-1} \sum 2|y - \hat{y}| / (|y| + |\hat{y}|)$, reported with its zero-demand distortions documented. Zero denominators stay honest NaNs ($n_{\text{NaN}} = 0$ everywhere); no epsilon guards.

Annexure D — Forecasting Results

The complete forecast boards are tables 4.1 and 4.2 (main text, Sections 4.5 and 4.6). The figure index below lists the main-text figures in narrative order. Main text FIG-01..FIG-28 in narrative order: intermittency quadrants, weekly seasonality, ACF/PACF, M5/Store representative series, Store diversity, M5 representativeness, model ladder, inventory connection, combined MASE board, full-ladder MAE, full-ladder WAPE, Croston MAE, archetype MAE, archetype WAPE, ARIMA-by-archetype, M5 cost-by-archetype, Store MAE (full500), Croston intuition, LSTM cell, total cost, service level, cost split, inventory levels, forecast-vs-inventory scatter, M5 rank heatmap, Store rank heatmap, winner-share. Full 97-figure audit with classes, placement, and captions: `data/figure_audit.csv`. Rejected: the three subset-SARIMA Store panels. Teaching figures beyond the four main-text selections are appendix-only or omitted.

Annexure E — Inventory Results

The complete inventory boards are tables 4.4 and 4.5 (main text, Sections [4.9](#) and [4.10](#)), with cost composition, service levels, and per-archetype figures in the same sections and the policy winner matrices in Annexure G.

Annexure F — Pairwise Comparisons

All 91 paired comparisons (55 M5 + 36 Store) with Holm-adjusted Wilcoxon p , paired d_z , and rank-biserial r . Generated from `06_results/statistical_tests/pairwise_tests.csv`; bootstrap CIs in the source file. Negative mean- d favors A.

Pairwise results: M5 (part 1 of 2).

Dataset	Pair (A vs. B)	Mean d	d_z	Holm p	r
M5	ARIMA vs. CROSTON	0.0503	0.036	5.48e-20	-0.028
M5	ARIMA vs. DES	-0.3815	-0.187	0.00e+00	-0.281
M5	ARIMA vs. LSTM	0.1433	0.119	3.56e-87	-0.060
M5	ARIMA vs. Moving Average	0.1327	0.115	9.36e-223	-0.102
M5	ARIMA vs. Naive	-0.0520	-0.073	2.59e-130	-0.100
M5	ARIMA vs. SBA	0.0659	0.057	4.44e-28	-0.033
M5	ARIMA vs. SES	0.1342	0.125	3.82e-127	-0.072
M5	ARIMA vs. Seasonal Naive	-0.0542	-0.034	1.50e-12	-0.027
M5	ARIMA vs. TES	-0.7898	-0.273	0.00e+00	-0.427
M5	ARIMA vs. TSB	0.1227	0.123	1.43e-72	-0.054
M5	CROSTON vs. DES	-0.4318	-0.170	0.00e+00	-0.191
M5	CROSTON vs. LSTM	0.0929	0.099	4.20e-227	-0.096
M5	CROSTON vs. Moving Average	0.0824	0.093	2.25e-118	-0.070
M5	CROSTON vs. Naive	-0.1023	-0.062	2.57e-38	-0.039
M5	CROSTON vs. SBA	0.0156	0.020	2.37e-63	-0.051
M5	CROSTON vs. SES	0.0839	0.101	1.27e-126	-0.072
M5	CROSTON vs. Seasonal Naive	-0.1045	-0.062	8.77e-32	-0.036
M5	CROSTON vs. TES	-0.8401	-0.239	0.00e+00	-0.258
M5	CROSTON vs. TSB	0.0724	0.078	1.49e-16	-0.025
M5	DES vs. LSTM	0.5248	0.213	0.00e+00	-0.228
M5	DES vs. Moving Average	0.5142	0.213	0.00e+00	-0.272
M5	DES vs. Naive	0.3296	0.148	0.00e+00	-0.217
M5	DES vs. SBA	0.4475	0.187	0.00e+00	-0.216
M5	DES vs. SES	0.5157	0.217	0.00e+00	-0.257
M5	DES vs. Seasonal Naive	0.3273	0.134	0.00e+00	-0.185
M5	DES vs. TES	-0.4083	-0.187	0.00e+00	-0.259
M5	DES vs. TSB	0.5042	0.216	0.00e+00	-0.276
M5	LSTM vs. Moving Average	-0.0105	-0.015	1.78e-03	-0.010

Pairwise results: M5 (part 2 of 2).

Dataset	Pair (A vs. B)	Mean d	d_z	Holm p	r
M5	LSTM vs. Naive	-0.1952	-0.136	9.82e-127	-0.072

Dataset	Pair (A vs. B)	Mean d	d_z	Holm p	r
M5	LSTM vs. SBA	-0.0773	-0.085	1.69e-86	-0.059
M5	LSTM vs. SES	-0.0090	-0.014	5.11e-04	-0.011
M5	LSTM vs. Seasonal Naive	-0.1975	-0.133	3.68e-124	-0.071
M5	LSTM vs. TES	-0.9331	-0.274	0.00e+00	-0.298
M5	LSTM vs. TSB	-0.0206	-0.031	2.02e-10	-0.020
M5	Moving Average vs. Naive	-0.1847	-0.129	8.28e-272	-0.114
M5	Moving Average vs. SBA	-0.0668	-0.073	4.20e-75	-0.055
M5	Moving Average vs. SES	0.0015	0.006	1.75e-12	-0.022
M5	Moving Average vs. Seasonal Naive	-0.1869	-0.129	6.33e-257	-0.111
M5	Moving Average vs. TES	-0.9225	-0.270	0.00e+00	-0.331
M5	Moving Average vs. TSB	-0.0100	-0.025	2.34e-43	-0.042
M5	Naive vs. SBA	0.1179	0.085	3.49e-54	-0.047
M5	Naive vs. SES	0.1862	0.136	2.74e-113	-0.068
M5	Naive vs. Seasonal Naive	-0.0022	-0.001	4.78e-01	-0.003
M5	Naive vs. TES	-0.7378	-0.248	0.00e+00	-0.410
M5	Naive vs. TSB	0.1746	0.132	8.08e-55	-0.047
M5	SBA vs. SES	0.0683	0.078	4.78e-63	-0.051
M5	SBA vs. Seasonal Naive	-0.1202	-0.076	5.89e-25	-0.031
M5	SBA vs. TES	-0.8558	-0.254	0.00e+00	-0.281
M5	SBA vs. TSB	0.0567	0.061	6.24e-23	-0.030
M5	SES vs. Seasonal Naive	-0.1884	-0.131	4.33e-92	-0.061
M5	SES vs. TES	-0.9240	-0.274	0.00e+00	-0.312
M5	SES vs. TSB	-0.0115	-0.041	5.08e-30	-0.035
M5	Seasonal Naive vs. TES	-0.7356	-0.224	0.00e+00	-0.334
M5	Seasonal Naive vs. TSB	0.1769	0.124	4.13e-42	-0.041
M5	TES vs. TSB	0.9125	0.274	0.00e+00	-0.330

Pairwise results: Store (36 pairs).

Dataset	Pair (A vs. B)	Mean d	d_z	Holm p	r
Store	ARIMA vs. DES	-1.7311	-0.138	4.34e-176	-0.085
Store	ARIMA vs. LSTM	4.0264	0.536	0.00e+00	-0.483
Store	ARIMA vs. Moving Average	2.1920	0.389	0.00e+00	-0.372
Store	ARIMA vs. Naive	-4.3916	-0.759	0.00e+00	-0.629
Store	ARIMA vs. SARIMA	3.4532	0.489	0.00e+00	-0.443
Store	ARIMA vs. SES	1.9780	0.411	0.00e+00	-0.385
Store	ARIMA vs. Seasonal Naive	2.4484	0.256	0.00e+00	-0.224
Store	ARIMA vs. TES	-4.9689	-0.299	0.00e+00	-0.261
Store	DES vs. LSTM	5.7575	0.477	0.00e+00	-0.444
Store	DES vs. Moving Average	3.9231	0.366	0.00e+00	-0.353

Dataset	Pair (A vs. B)	Mean d	d_z	Holm p	r
Store	DES vs. Naive	-2.6605	-0.172	0.00e+00	-0.228
Store	DES vs. SARIMA	5.1843	0.463	0.00e+00	-0.429
Store	DES vs. SES	3.7091	0.311	0.00e+00	-0.294
Store	DES vs. Seasonal Naive	4.1795	0.338	0.00e+00	-0.300
Store	DES vs. TES	-3.2378	-0.254	0.00e+00	-0.234
Store	LSTM vs. Moving Average	-1.8345	-0.295	0.00e+00	-0.282
Store	LSTM vs. Naive	-8.4180	-0.831	0.00e+00	-0.694
Store	LSTM vs. SARIMA	-0.5732	-0.118	0.00e+00	-0.128
Store	LSTM vs. SES	-2.0484	-0.336	0.00e+00	-0.311
Store	LSTM vs. Seasonal Naive	-1.5781	-0.251	0.00e+00	-0.248
Store	LSTM vs. TES	-8.9953	-0.567	0.00e+00	-0.542
Store	Moving Average vs. Naive	-6.5836	-0.684	0.00e+00	-0.589
Store	Moving Average vs. SARIMA	1.2612	0.184	0.00e+00	-0.170
Store	Moving Average vs. SES	-0.2139	-0.082	5.22e-198	-0.090
Store	Moving Average vs. Seasonal Naive	0.2564	0.031	9.82e-12	-0.020
Store	Moving Average vs. TES	-7.1608	-0.442	0.00e+00	-0.415
Store	Naive vs. SARIMA	7.8448	0.757	0.00e+00	-0.644
Store	Naive vs. SES	6.3696	0.751	0.00e+00	-0.629
Store	Naive vs. Seasonal Naive	6.8400	0.581	0.00e+00	-0.581
Store	Naive vs. TES	-0.5773	-0.031	2.25e-19	-0.027
Store	SARIMA vs. SES	-1.4752	-0.206	0.00e+00	-0.183
Store	SARIMA vs. Seasonal Naive	-1.0048	-0.148	0.00e+00	-0.141
Store	SARIMA vs. TES	-8.4221	-0.567	0.00e+00	-0.550
Store	SES vs. Seasonal Naive	0.4703	0.056	2.59e-32	-0.036
Store	SES vs. TES	-6.9469	-0.421	0.00e+00	-0.394
Store	Seasonal Naive vs. TES	-7.4172	-0.464	0.00e+00	-0.430

Annexure G — Sensitivity Analysis

Winner per policy cell ($H = 1$ throughout). M5: LSTM everywhere except two SES cells at short lead/high penalty. Store: short leads → Moving Average/DES; mid leads → SES/Moving Average; long leads → Naive/SES; LSTM wins no Store cell — the fragility verdict in tabular form. Full 540-row costs: `06_results/sensitivity/sensitivity_grid.csv`.

Policy winners: M5 (27 cells).

Dataset	L	Service	P	Winner
M5	3	0.90	3	LSTM
M5	3	0.90	5	LSTM
M5	3	0.90	10	SES
M5	3	0.95	3	LSTM
M5	3	0.95	5	LSTM
M5	3	0.95	10	SES
M5	3	0.99	3	LSTM
M5	3	0.99	5	LSTM
M5	3	0.99	10	LSTM
M5	7	0.90	3	LSTM
M5	7	0.90	5	LSTM
M5	7	0.90	10	LSTM
M5	7	0.95	3	LSTM
M5	7	0.95	5	LSTM
M5	7	0.95	10	LSTM
M5	7	0.99	3	LSTM
M5	7	0.99	5	LSTM
M5	7	0.99	10	LSTM
M5	14	0.90	3	LSTM
M5	14	0.90	5	LSTM
M5	14	0.90	10	LSTM
M5	14	0.95	3	LSTM
M5	14	0.95	5	LSTM
M5	14	0.95	10	LSTM
M5	14	0.99	3	LSTM
M5	14	0.99	5	LSTM
M5	14	0.99	10	LSTM

Policy winners: Store (27 cells).

Dataset	L	Service	P	Winner
Store	3	0.90	3	Moving Average

Dataset	L	Service	P	Winner
Store	3	0.90	5	DES
Store	3	0.90	10	DES
Store	3	0.95	3	Moving Average
Store	3	0.95	5	DES
Store	3	0.95	10	DES
Store	3	0.99	3	Moving Average
Store	3	0.99	5	Moving Average
Store	3	0.99	10	DES
Store	7	0.90	3	SES
Store	7	0.90	5	Moving Average
Store	7	0.90	10	Moving Average
Store	7	0.95	3	SES
Store	7	0.95	5	Moving Average
Store	7	0.95	10	Moving Average
Store	7	0.99	3	SES
Store	7	0.99	5	SES
Store	7	0.99	10	Moving Average
Store	14	0.90	3	Naive
Store	14	0.90	5	Naive
Store	14	0.90	10	SES
Store	14	0.95	3	Naive
Store	14	0.95	5	Naive
Store	14	0.95	10	SES
Store	14	0.99	3	Naive
Store	14	0.99	5	Naive
Store	14	0.99	10	SES

Annexure H — Convergence and Validation

The independent validator (`11_src/validate_results.py`) enforces 44 gates: per-family row counts (112,000), zero duplicate keys, origins 1–8, horizon 28 everywhere, test-window containment, finite non-negative forecasts, cross-family actuals identity (spread 0.0), metric presence with zero unexpected NaNs, and SARIMA-full500 presence (112,000 rows). Result: 44 PASS, 0 FAIL (`06_results/validation_report.json`). Gate log:

Validation gates (part 1).

Gate	Status	Detail
sarima full500 present	PASS	112000 rows
baselines row counts	PASS	(('store_item_demand', 'Seasonal Naive')):
baselines no dup keys	PASS	dups=0
baselines origins 1 to 8	PASS	[np.int64(1), np.int64(2), np.int64(3), np.i
baselines h28 everywhere	PASS	min=28 max=28
baselines dates in test window	PASS	
baselines forecasts finite nonneg	PASS	
exponential smoothing row counts	PASS	(('store_item_demand', 'TES'): 112000, ('m
exponential smoothing no dup keys	PASS	dups=0
exponential smoothing origins 1 to 8	PASS	[np.int64(1), np.int64(2), np.int64(3), np.i
exponential smoothing h28 everywhere	PASS	min=28 max=28
exponential smoothing dates in test window	PASS	
exponential smoothing forecasts finite nonneg	PASS	
arima row counts	PASS	(('store_item_demand', 'ARIMA'): 112000, (
arima no dup keys	PASS	dups=0
arima origins 1 to 8	PASS	[np.int64(1), np.int64(2), np.int64(3), np.i
arima h28 everywhere	PASS	min=28 max=28
arima dates in test window	PASS	
arima forecasts finite nonneg	PASS	
lstm row counts	PASS	(('store_item_demand', 'LSTM'): 112000, ('
lstm no dup keys	PASS	dups=0
lstm origins 1 to 8	PASS	[np.int64(1), np.int64(2), np.int64(3), np.i

Validation gates (part 2).

Gate	Status	Detail
lstm h28 everywhere	PASS	min=28 max=28
lstm dates in test window	PASS	
lstm forecasts finite nonneg	PASS	
croston row counts	PASS	((('m5', 'SBA'): 112000, ('m5', 'CROSTON'): 1

Gate	Status	Detail
croston no dup keys	PASS	dups=0
croston origins 1 to 8	PASS	[np.int64(1), np.int64(2), np.int64(3), np.i
croston h28 everywhere	PASS	min=28 max=28
croston dates in test window	PASS	
croston forecasts finite nonneg	PASS	
cross family actuals identical	PASS	max spread=0.000000
baselines metrics have MASE RMSSE	PASS	['dataset', 'model', 'MAE', 'RMSE', 'sMAPE',
baselines metrics no unexpected NaN	PASS	Empty DataFrame Columns: [dataset, model] In
exponential smoothing metrics have MASE RMSSE	PASS	['dataset', 'model', 'MAE', 'RMSE', 'sMAPE',
exponential smoothing metrics no unexpected NaN	PASS	Empty DataFrame Columns: [dataset, model] In
arima metrics have MASE RMSSE	PASS	['dataset', 'model', 'n', 'MAE', 'RMSE', 'sM
arima metrics no unexpected NaN	PASS	Empty DataFrame Columns: [dataset, model] In
lstm metrics have MASE RMSSE	PASS	['dataset', 'model', 'MAE', 'RMSE', 'sMAPE',
lstm metrics no unexpected NaN	PASS	Empty DataFrame Columns: [dataset, model] In
croston metrics have MASE RMSSE	PASS	['dataset', 'model', 'MAE', 'RMSE', 'sMAPE',
croston metrics no unexpected NaN	PASS	Empty DataFrame Columns: [dataset, model] In
inventory covers all models	PASS	missing=set() extra=set()
inventory costs finite	PASS	

Per-fit logs: `convergence_details_all.csv` (ARIMA both datasets),
`convergence_details_full1500.csv` (4,000 SARIMA fits, all `fit_ok`),
`run_log_full1500.json` (seed 42, 2,382 s, 0 skips, 0 failures), `sarima_feasibility.json` (16-
fit probe, 0.62 s/fit mean). M5 ARIMA's 906 pre-check fallbacks concentrate in Highly Intermittent
series per the per-fit archetype field. Scope column (`subset100_exploratory`
vs. `full1500_primary`) separates the exploratory subset from the primary run everywhere.

Annexure I — Reproducibility and Provenance

Repository root: AI-INVENTORY-OPTIMIZATION, branches main. Evidence-freeze commits: 53c8343 (final results), 7da7c45 (generators), edb42ad (tracked asset package). Seeds: 42 in config, notebooks, torch/numpy, SARIMA scripts, validator sampling. Manifests: smoothing execution metadata (121 d), Croston validation selection (121 d + seed), SARIMA run log, feasibility probe, validation report (0 FAIL), pre-hardening backup manifest. Pre-hardening originals: 06_results/_pre_hardening_backup/ (10 metric files + subset-based forecasts) — historical, never report sources. Stale/self-excluded: 06_results/inventory/phase1_*.csv (legacy names), 12_notebooks_pdf/ (pre-rerun export), subset-SARIMA Store figure panels. Report inputs: 09_reports/final/ (README, number sheet, audits, traceability, staged figures, rebuild generators 11_src/build_number_sheet.py, 11_src/rebuild_report_figures.py, 11_src/build_figure_audit.py, 11_src/gen_appendix_tables.py).

00 research map (framing); 01 environment/reproducibility; 02a M5 acquisition/audit; 02b Store acquisition/audit; 03 EDA (M5 deep dive); 04 series selection/experimental design (frozen numbers); 05a time-series components (teaching); 05b baselines (re-executed 2026-09-03 15:35); 06 smoothing (15:36–37); 06c Croston (15:42); 07 ARIMA/SARIMA (15:55 + full500 script 22:49); 08 LSTM (hardened rerun 18:00). Full purpose/figure/table/section mapping: traceability/notebook_traceability.csv.

Annexure J — Supporting Tables and Figures

Table audit (T-01..T-09 + appendix tables with exact sources): data/table_audit.csv. Number sheet (188 rows, programmatic): data/final_number_sheet.csv. Claim-to-source matrix (14 claims): traceability/claim_traceability.csv. Key lookup values: M5 MASE board LSTM 1.316 / MA 1.346 / SES 1.351 / TSB 1.369; Store MASE LSTM 0.978 / SARIMA-full500 1.055 / SNaive 1.183; M5 costs LSTM 152.83 through TES 346.68; Store costs MA 2,084.50 through TES 3,419.88; policy wins M5 LSTM 25/SES 2, Store MA 9/SES 7/Naive 6/DES 5; convergence M5 ARIMA 22.65% pre-check, Store 0%, SARIMA-full500 4,000/4,000.

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